OXFORD UNIVERSITY PRESS | Annals of Botany

Plants in limbo — only macroalgal detritus can remain viable for months

Journal:	*Annals of Botany*
Manuscript ID	Draft
Manuscript Type:	Research in Context
Date Submitted by the Author:	n/a
Complete List of Authors:	Wright, Luka; UWA Oceans Institute, ; The University of Western Australia School of Biological Sciences,
Subject Area:	angiosperm evolution, Bayesian modelling and statistics, big data, macroalgae, marine botany, leaf photosynthesis, seagrass, seaweeds
Subject Area (free entry):	

SPECIAL ISSUE: PLANT SENESCENCE

RESEARCH IN CONTEXT

Plants in limbo — o~~O~~nly macroalgal detritus can remain viable for months

Luka Seamus Wright[1,2]*

[1]Oceans Institute, University of Western Australia, Perth, Australia

[2]School of Biological Sciences, University of Western Australia, Perth, Australia

*luka.wright@research.uwa.edu.au, luka@wright.it

Running title

Only macroalgal detritus can remain viable for months

Abstract

Background and Aims Seaweed detritus, particularly that of kelps, can maintain photosynthesis over several months following detachment. By delaying and even counteracting decomposition, detrital photosynthesis may have a substantial effect on detrital dynamics and thus carbon cycling. Viability of detritus could be explained by limited tissue differentiation, as is common in non-vascular plants. However, it remains unclear if detrital photosynthesis is restricted to this group. Here I look for detrital photosynthesis in seagrasses for which there is data scarcity and then compare them to seaweeds and other plants *sensu lato* to determine the relative influence of plant physiology on decomposition.

Methods I excised leaves of the seagrasses *Amphibolis antarctica* and *Halophila ovalis* by cutting the petiole, mimicking detachment caused by hydrodynamics or herbivory which prematurely induces senescence. Leaves were weighed down on sediment in mesh bags, supplied with fresh flowing seawater and light, and periodically destructively sampled to measure light-saturated net photosynthesis in closed oxygen incubations. To view my research in context, I performed a meta-analysis of detrital photosynthesis and chlorophyll against time post-excision across >10^4 observations from 127 independent studies on 92 species from 37 families and 23 orders of terrestrial and aquatic plants.

Key Results Here I show that detrital photosynthesis *per se* is not restricted to non-vascular plants. Seagrasses exhibit detrital photosynthesis, but only up to a month post-excision. This is substantially longer than for terrestrial and freshwater plants (one week) but also much shorter than for seaweeds (several months, possibly up to a year).

Conclusions Detrital photosynthesis is clearly a phenomenon beyond the realm of seaweeds. The intermediate longevity of seagrass detritus is likely due to convergent evolution with seaweeds. However, months-long detrital photosynthesis is unique to macroalgae. So physiology probably only substantially influences detrital recalcitrance in this plant group. My findings call into question our grasp of the key predictors of seaweed decomposition and thereby of blue carbon in general.

Keywords

Laminariales, Alismatales, Hydrocharitaceae, Cymodoceaceae, paddle weed, wire weed, leaf litter, drift, dislodgement, abscission, degradation, decay.

Introduction

…tossed and swept like dead leaves from one spot to another, never resting, never giving back their goodness to the earth, never fully dead but in some vast limbo between life and extinction, tossing and tumbling without end… ~~—~~—Philip Pullman (1994) *The Tin Princess*

Seaweed detrital dynamics are strongly influenced by physiology. Except for wrack, k~~K~~elp detritus—*sensu lato* tissue that becomes detached—~~—~~can remain physiologically viable for months (de Bettignies *et al.*, 2020; Frontier *et al.*, 2021; Wright and Foggo, 2021; Wright *et al.*, 2022; Wright and Kregting, 2023; Wright *et al.*, 2024), often resulting in atypical decomposition trajectories (Kennedy and Blain, 2025). Floating kelp detritus sinks before it stops photosynthesising (Graiff *et al.*, 2013, 2016; Tala *et al.*, 2019). And if the fragment happens to be reproductive tissue, detritus can even be a vector for dispersal (Macaya *et al.*, 2005; Fraser *et al.*, 2018; Tala *et al.*, 2019; de Bettignies *et al.*, 2020). This stands in stark contrast to the traditional ecological definition of detritus as "dead organic matter" (Moore *et al.*, 2004, cf. Wright and Kregting, 2023). There is no word for plant material that is detached but living and since detached seaweed—except for holopelagic species—has been referred to as detritus or less commonly as drift or litter all along, I propose to continue using detritus for any detached tissue and detritus *sensu stricto* for detached and dead tissue. Detrital photosynthesis provides the strongest evidence in support of the viability of detritus. But the fundamental impact on detrital dynamics only becomes apparent when comparing decomposition rates between live and dead or senescent detritus. ~~In the absence of physiology~~When it is dead, seaweed detritus decomposes at least twice as fast (Birch *et al.*, 1983; Brouwer, 1996; de Bettignies *et al.*, 2020; Smith and Foreman, 1984 vs. Albright *et al.*, 1982, Bedford and Moore, 1984). This is likely due to maintenance of growth and chemical defence (Tala *et al.*, 2019; de Bettignies *et al.*, 2020;

Frontier *et al.*, 2021; Wright *et al.*, 2022), probably making physiology an important source of detrital recalcitrance (Wright *et al.*, 2022, 2024).

There is currently no baseline for detrital photosynthesis—the persistence of photosynthesis in the detrital phase. The term was first introduced quite recently in the context of kelp decomposition (Frontier *et al.*, 2021; Wright *et al.*, 2022) although there was clearly awareness of the phenomenon in much earlier phycology research which deemed it necessary to pre-kill detritus (Birch *et al.*, 1983; Smith and Foreman, 1984; Brouwer, 1996). Other fields of botany do not use the term but there certainly has been abundant research into physiology of excised leaves of terrestrial and freshwater plants (e.g. Lovell *et al.*, 1972; Thimann and Satler, 1979; Jana and Choudhuri, 1980; Kar and Choudhuri, 1986). The literature also includes some discussion on the ~~meaning~~ absence of chlorophyll ~~persistence~~ degradation in senescing seagrass (Knauer and Ayers, 1977; Pellikaan, 1982; Strother and Vatta, 1986). Yet no attempt has been made to align these seemingly incongruous research trajectories in a formal comparison. We consequently don't know if detrital photosynthesis is as unique as it seems. Instead, it may be quite prevalent among plants. A reviewer once wrote to me that detrital photosynthesis in kelps "is really not surprising, since macroalgae are much less differentiated than higher plants and, thus, there is no reason why the single cell shouldn't be able to continue its activity for a while". To satisfy their mentality I added a statement to the effect that detrital photosynthesis is "readily explained" by the ability of all macroalgal tissues to independently photosynthesise to some extent (Wright and Kregting, 2023). In truth, nothing is "readily explained" due to our lack of reference points. The mechanism underlying the ability of seaweeds to maintain photosynthesis in the detrital phase remains unclear. It could be their aquatic nature which removes reliance on roots for water, nutrient provisioning from the surrounding water, non-vascularity, minimal tissue differentiation and/or something altogether different. To understand the prerequisite for detrital photosynthesis and thus predict prevalence among plants, model systems beyond seaweeds are needed.

Seagrasses are suitable candidates for further investigation of detrital photosynthesis. While also marine macrophytes, they are fundamentally different from seaweeds. Seagrasses are vascular plants which, due to their terrestrial ancestry (Olsen *et al.*, 2016), are differentiated into roots, rhizomes or stems, and leaves. However, convergent evolution with seaweeds (Olsen *et al.*, 2016) has reduced their reliance on roots for nutrient uptake. Seagrass leaves tend to outperform their roots in suppling nutrients to the plant (Pedersen and Borum, 1992; Stapel *et al.*, 1996; Pedersen *et al.*, 1997; Gras *et al.*, 2003; Viana *et al.*, 2019). This suggests that seagrass leaves can independently photosynthesise, seemingly reducing the primary function of the roots to anchorage, analogous to the holdfast of seaweeds. However, given the usually greater ammonium concentration in sediment pore water, roots can supply more nitrogen despite the leaves' greater uptake affinity (Short and McRoy, 1984; Lee and Dunton, 1999, but see Pedersen and Borum, 1992) and nutrients are translocated to leaves (Short and McRoy, 1984; Viana *et al.*, 2019), even those tens of centimetres from the source (Marbà *et al.*, 2002). Roots can also tap into nutrient pools that are unavailable to leaves, such as particulate (Evrard *et al.*, 2005) and microbial (Patriquin and Knowles, 1972; Mohr *et al.*, 2021) nitrogen. Clearly the photosynthetic emancipation of seagrass leaves is at a halfway point between terrestrial plants and macroalgae, so seagrasses probably display detrital photosynthesis to some extent. Relatively slow and light-dependent chlorophyll degradation in seagrass detritus (Knauer and Ayers, 1977; Pellikaan, 1982; Strother and Vatta, 1986) provides some support for this, but there are currently no data on seagrass detrital photosynthesis. Freshwater and terrestrial plants are expected to exhibit detrital photosynthesis to a similar and lesser extent respectively compared to seagrasses.

Here I aim to ~~verify the existence of~~quantify detrital photosynthesis in seagrasses and ~~test its extent in~~ plants more broadly to ~~determine~~ predict ~~potential influences on~~the extent to which it may influence plant detrital dynamics. I hope this information will allow us to re-evaluate seaweed and seagrass decomposition in the context of plants at large. To address these aims I conducted a brief *ex situ* decomposition experiment with the two diverse seagrasses *Halophila ovalis* (Hydrocharitaceae) and *Amphibolis antarctica* (Cymodoceaceae). Apart from their phylogenetic (diverged 105 Ma ago, Waycott *et al.*, 2018) and obvious morphological (de los Santos *et al.*, 2012, 2016) differences, I chose these genera because they give some indication of the possibility of detached photosynthesis (Kenworthy *et al.*, 1989; Pedersen *et al.*, 1997). I excised fresh leaves and left them to decompose on sediment, measuring net photosynthesis of destructively sampled leaves at regular intervals. Excision was intended to mimic detachment of non-senescent leaves rather than abscission. I then compared my evidence with published data on seaweeds as well as terrestrial and freshwater plants by pulling together various lines of research in a single meta-analysis.

Materials and methods

Experiment (the research)

Halophila ovalis ramets with several leaf pairs and intact roots were collected from Pelican Point in the brackish Swan Estuary (31.98715°S, 115.82101°E) alongside sediment from adjacent Matilda Bay (31.97572°S, 115.82324°E) on 15th September 2022. I filled the bottom of a 220-L glass flow-through seawater tank with the sediment and planted the ramets to ~~briefly~~ acclimatise them to ambient seawater temperature and salinity and a 12-12-h light-dark cycle (11 ± 1.6 µmol photons m^{-2} s^{-1}, mean ± s.d., cf. Masini and Manning, 1997)~~)~~ for about two hours. *Amphibolis antarctica* stems with several leaf clusters were collected from fully marine Marmion Lagoon (31.85303°S, 115.75092°E) on 13th October 2022 and immediately used in the experiment. Sediment and seagrasses were collected from less than one metre depth.

The *ex situ* decomposition experiment was commenced on the respective day of collection by excising individual leaves (*H. ovalis*) or leaf clusters (*A. antarctica*), hereinafter leaves, and immediately measuring photosynthesis (see next paragraph) of six (*H. ovalis*) or nine (*A. antarctica*) samples. The remaining leaves were enclosed in mesh (3-mm ⌀). Mesh bags were weighed down with inert aquarium pebbles and placed on the sediment in the same 220-L tank under identical photon fluence rate of 11 ± 1.6 µmol photons m^{-2} s^{-1}, which is more than sufficient for detrital photosynthesis to play a role (Wright *et al.*, 2024) and is representative of the light environment experienced by detritus in the understorey of kelp forests and seagrass meadows or on deeper sediment flats. To measure photosynthesis a~~A~~t each of five (*A. antarctica*) to six (*H. ovalis*) timepoints over 34 d following excision, six destructive samples were taken from the mesh bags which were kept submerged for the entire duration of the experiment ~~to measure photosynthesis~~.

Light-saturated net photosynthesis (P_{max}) of leaves was measured using closed oxygen (O_2) incubations (c~~.~~f. Wright *et al.*, 2024). *A. antarctica* leaf clusters weigh roughly ten times as much as *H. ovalis* leaves (in light of my data the mass ratio seems to be closer to 14:1). Since I wanted to measure both seagrasses ~~in~~ at ~~the same~~a comparable leaf mass per volume for the same duration, I incubated ten *H. ovalis* leaves or single *A. antarctica* leaf clusters alongside seawater blanks in sealed glass jars filled with water collected from the flow-through system. Each jar was equipped with a magnetic stir bar and a self-adhesive planar O_2 sensor spot (SP-PSt3-SA-NAU-D5-YOP, PreSens Precision Sensing GmbH, Regensburg, Germany) and placed on a magnetic stirrer under a saturating (Masini and Manning, 1997; Jamaludin *et al.*, 2006; Said *et al.*, 2021) irradiance of 420 ± 19 µmol photons m^{-2} s^{-1} (Zeus, Ledzeal, Shenzhen Topline Lighting Technology Co. Ltd., Shenzhen, China) in a 20ºC room. Care was taken to exclude air from jars. Dissolved O_2 (µM) was measured fibre-optically through the glass every 10 s over at least 35 min with a four-channel O_2 meter (OXY-4 SMA G2, PreSens Precision Sensing GmbH, Regensburg, Germany). The O_2 meter was calibrated using anoxic (1% w/v Na_2SO_3) and air-saturated (bubbled with air) ultrapure water. All measurements were corrected for incubation temperature (18 ± 0.73 ºC), pressure (1020 ± 6 hPa) and salinity (35 ± 0.48 ‰) using a single temperature dipping probe connected to the first channel of the O_2 meter's built-in temperature sensor and placed in a fifth jar filled with seawater, the O_2 meter's built-in pressure sensor and a handheld refractometer respectively. I blotted and weighed (0.01-g accuracy, 440-33N, Kern & Sohn GmbH, Balingen, Germany) leaves and gravimetrically determined jar volume with ultrapure water to standardise P_{max} as shown in the Equation 1.

Data analysis and visualisation were performed in R v4.2.3 (R Core Team, 2025) with the tidyverse package family v2.0.0 (Wickham *et al.*, 2019) within the integrated development environment RStudio v2023.06.0+421 (RStudio Team, 2025). Hamiltonian Monte Carlo models were written in Stan (Carpenter *et al.*, 2017) and run with the R interface cmdstanr v0.5.3 (Gabry *et al.*, 2024) via CmdStan v2.30.1 (Lee *et al.*, 2017). All models were run with 8 Markov chains spread across all cores with 10^4 warmup and sampling iterations each. Convergence and smooth sampling were optimised by assessing effective sample sizes and $\hat{R}$ scores and visually scrutinising trace rank and pair plots with bayesplot v1.11.1 (Gabry *et al.*, 2019). All reported results are posterior probabilities and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay, 2024). The R script can be consulted for detailed information on data analysis (github.com/lukaseamus/~~seagrass-detrital-photosynthesis~~plant-limbo). Vector illustrations were made in Affinity Designer v1.10.6 (Serif Ltd., Nottingham, UK).

In the first instance, simple linear models with centred incubation time ($t - \bar{t}$, min) as the predictor and dissolved O_2 (µM) as the response variable were fit to measurements from each sample and blank incubation. This yielded posterior probability distributions for slopes (β, µM min^{-1}) which were converted to mass-based P_{max} (µmol g^{-1} h^{-1}) as

$$P_{\max} = \frac{(\beta_s -- \beta_b) \times V \times \Delta t}{m} \tag{1}$$

where subscript s and b denote sample and blank incubations from the same measurement group, V is the incubation volume (175 ± 0.17 mL) in litres, m is the sample blotted mass (0.52 ± 0.23 g) in grams and Δt is the desired period (60 min h^{-1}) in minutes.

To predict P_{max} with detrital age (A, d), I chose a logistic regression of the form

$$P_{\max} \sim N(P_\mu, P_\sigma) \tag{2}$$

$$P_{\mu} = \frac{\alpha + \tau}{1 + e^{k \times (A - \mu)}} - \tau$$

where α is baseline photosynthesis (µmol g^{-1} h^{-1}), τ is the ~~community~~ oxygen consumption of dead detritus (µmol g^{-1} h^{-1}), k is the logistic rate of decay (d^{-1}) and μ is the midpoint of photosynthetic death, i.e. half-life of photosynthesis (d). The naming of parameters is inspired by the Hebrew word אמת (graecised τµα) which means truth and is composed of the first, middle and last letters of the Hebrew alphabet, symbolising life (א), transition (מ) and death (ת). Inscribed on the golem's brow this word lends life, but removing א changes the meaning from truth to death (מת) and the golem dies. As in the story of the golem, the logistic function is optimal for modelling a shift between two alternate states and is thus the logical choice here. I previously modelled detrital photosynthesis with linear regressions instead, because data either did not span enough timepoints to inform a more complex model (Wright *et al.*, 2022) or macroalgal tissue disintegration coinciding with photosynthetic death prevented estimation of μ and τ (Wright *et al.*, 2024). However, in the present case these limitations do not apply, since I collected data at six to seven timepoints and seagrass leaves do not disintegrate at or shortly after photosynthetic death, allowing me to choose the optimal model. The downside of using more complex models is that multiple regression to account for technical confounders (Wright *et al.*, 2024) is not straightforward. So I opted for a separate multiple linear regression of P_{max} against the standardised potential confounders initial incubation O_2 (µM), mean incubation temperature (°C), mean incubation pressure (hPA), incubation salinity (‰) and leaf blotted mass (g) (Figure S1).

Prior probability distributions for parameters were chosen based on published estimates and logic. For the linear regressions of O_2 against incubation time, I gave the intercept a normal prior centred on regional average seawater O_2 and the slope a normal prior with a mean of zero. For a (Equation 2), I chose a gamma prior informed by various published P_{max} estimates for *H. ovalis* (40 ± 57 µmol g^{-1} h^{-1}, 1.8 ± 2.5 µmol $leaf^{-1}$ h^{-1}, n = 32, Björk *et al.*, 1997; Jamaludin *et al.*, 2006; Borum *et al.*, 2016; Lamit and Tanaka, 2021; Said *et al.*, 2021, 2024) and *A. antarctica* (14 ± 9 µmol g^{-1} h^{-1}, 8.8 ± 5.5 µmol $leaf^{-1}$ h^{-1}, n = 8, Masini and Manning, 1997; Borum *et al.*, 2016; Said *et al.*, 2024). Units were converted where necessary by multiplying by mean dry-fresh mass ratios for *H. ovalis* (0.21 g dry mass g^{-1} fresh mass) and *A. antarctica* (0.29 g dry mass g^{-1} fresh mass) (de los Santos *et al.*, 2012; Borum *et al.*, 2016) and my own leaf fresh masses for *H. ovalis* (0.04 ± 0.01 g $leaf^{-1}$) and *A. antarctica* (0.61 ± 0.27 g $leaf^{-1}$). For τ (Equation 2), I chose a gamma prior to logically rule out net photosynthesis of dead detritus and centred it around published estimates of O_2 consumption by litter of *Zostera marina* (17 ± 5.6 µmol g^{-1} h^{-1}, n = 16, Blum and Mills, 1991) and *Posidonia oceanica* (1.3 ± 1.3 µmol g^{-1} h^{-1}, n = 46, Mateo and Romero, 1996, 1997). Units were again converted by multiplying by mean dry-fresh mass ratios for *Z. marina* (0.23 g dry mass g^{-1} fresh mass, Evans *et al.*, 1986) and *P. oceanica* (0.24 g dry mass g^{-1} fresh mass, Apostolaki *et al.*, 2024). I decided k (Equation 2) would best be restricted to positive values to ensure photosynthetic decay and would likely be higher than for seaweeds, so chose a gamma prior with a mean of 0.2 d^{-1}, the maximum for *Ecklonia radiata* (Wright *et al.*, 2024). For μ (Equation 2), since detrital age is inherently positive, I settled on a gamma prior with a mean of 17 d, half the experimental duration. I chose a reasonable s.d. for each prior distribution based on published estimates and prior simulation. Importantly, to enforce an y~~x~~-intercept close to maximum~~1~~, I ~~put an additional joint prior~~

~~on~~parameterised the model in terms of μ and the ~~k and μ which favoured a~~ logistic intercept ($k \times \mu$, log odds at $A = 0$) for which I chose a prior centred on ~~close to 4 ± 1~~5, ~~or~~ translating to about ~~0.97 ± 0.024 on the probability scale~~99% of the maximum value. ~~This was coded as `target += gamma_lpdf( k .* mu | 4^2 / 1^2 , 4 / 1^2 )` in Stan (Carpenter *et al.*, 2017).~~ Priors are visualised alongside posteriors for scrutiny.

Finally, the treatment of uncertainty deserves brief mention. Firstly, I propagated measurement error. Specifically, β and V (Equation 1) as well as initial incubation O_2 and incubation temperature are measured with error, which I incorporated into the downstream models as s.d. of posterior distributions. Wherever a variable is measured with error, I visualised each observation as a distribution rather than a point. Secondly, I applied partial pooling to the species variable to estimate uncertainty within and across species and make predictions for new seagrasses.

Meta-analysis (the context)

I searched the literature using keyword strings such as "(photosynthesis OR chlorophyll) AND ('induced senescence' OR detrit* OR detach* OR excis* OR cut OR isolate*) AND (week* OR day* OR hour* OR minute* OR time OR age)" in Web of Science (webofscience.com/wos) and Google Scholar (scholar.google.com). Once I had a selection of suitable papers, I used Crossref Metadata Search (search.crossref.org) to find similar papers. I accepted papers on all plants *sensu lato* (Bolton, 2016) that induced senescence by excision or detachment at an exact timepoint and repeatedly measured ~~various photosynthesis response variables~~photosynthesis to assess viability of the excised or detached tissue over any interval.~~,~~ Studies on holopelagic plants, such as *Sargassum fluitans* and *S. natans*, and abscission were not considered since the predictor variable detrital age cannot be exactly determined, either because the plant is detached to begin with or senescence started at an unknown time prior to detachment. I accepted various photosynthesis response variables, including O_2 and CO_2 gas exchange, carbon assimilation (^{14}C concentration), chlorophyll fluorescence (F_v/F_m), photosystem I and II electron transport rates (Mehler and Hill reactions), RuBisCo activity or concentration, carbonic anhydrase activity and photorespiration (glycolate concentration), as well as total chlorophyll (Chl), Chl *a* or a chlorophyll indicator. Microbial photosynthesis was generally assumed to be negligible since most of the mentioned response variables would likely not detect it. Regardless, they should be considered measures of holobiont photophysiology. This resulted in 127 suitable independent studies between 1957 and the present study (see supplementary references), from which I extracted data by copying tables or digitising plots using WebPlotDigitizer v5.2 (Rohatgi, 2025). In two cases I contacted authors for their raw data and in five cases the data were my own. It is noteworthy that apart from these seven cases, raw data were not available.

Data were collated into variables Reference, DOI, Group, Phylum, Order, Family, Species, Light, Water, Series, Day, Mean, SEM, N, Response, Method, Unit and Source. Some of these are self-explanatory, but most are not. Group classes observations into four non-taxonomic groups of interest: terrestrial plants (66 species), freshwater plants (6 species), seagrasses (5 species) and seaweeds (15 species). It is noteworthy that all freshwater plants belonged to the order Alismatales, like seagrasses, and all seaweeds belonged to the green and brown algae (Table S1). My distinction between freshwater and terrestrial plants is based on whether or not the leaves are submerged. There were no studies on freshwater macroalgae, so seaweeds are presumed to be representative. Phylum, Order, Family and Species represent the currently accepted taxonomy according to Plants of the World Online (POWO, 2025) and AlgaeBase (Guiry and Guiry, 2025). Light and Water are binary classifiers of whether or not the experimental plant tissue had access to light or water. Series numbers the measurement timeseries within a given study since most studies reported several experiments, response variables, species or individuals. Only measurement series with at least three timepoints were accepted. Day is the time post-excision given in d ($\frac{\mathrm{wk}}{7}$, h × 24, min × 1440). Mean is either an observation or the mean of several observations at a given timepoint. When the case is the latter, SEM and N are the standard error of the mean (s.e.m.) and its sample size (n). Uncertainty was mostly given as s.e.m., but when s.d. was provided instead, this was converted as $\mathrm{s.e.m.} = \frac{\mathrm{s.d.}}{\sqrt{\mathrm{n}}}$. In a few cases uncertainty was given as a 95% confidence interval (CI) or interquartile range (IQR). In the former case, I converted as $\mathrm{s.e.m.} = \frac{0.5 \times \mathrm{CI}_{0.95}}{\Phi^{-1}(0.975)}$, in the latter I assumed mean = median and conservatively converted the larger of the two quartile ranges ($Q_3 - Q_2$ or $Q_2 - Q_1$) as $\mathrm{s.e.m.} = \frac{\mathrm{QR}}{\Phi^{-1}(0.75) \times \sqrt{\mathrm{n}}}$. Response dichotomises data into photosynthesis and chlorophyll measurements since these measures are decoupled and probably senesce on different timescales. Method details how the response variable was measured, Unit provides the original response unit, and Source directs the reader to the data source in the paper. Please refer to the data publication (Wright, 2025) and github.com/lukaseamus/detrital-photosynthesis for further details.

The resulting 535 measurement series contained a mixture of observations and means. To jointly analyse data with such different levels of uncertainty one must either condense observations to means and s.e.m. or s.d. and build a measurement error model or expand means to observations. I opted for the latter and obtained 10566 observations by simulating draws from the normal distribution using `rnorm( N , Mean , SEM * sqrt(N) )` in R (R Core Team, 2025), or mathematically $X_{1:N} \sim \mathrm{N}(\,\mathrm{Mean},\ (\mathrm{SEM} \times \sqrt{\mathrm{N}})^2\,)$. Prior to analysis, I normalised observations to proportions by dividing by the initial observation or mean of initial observations in the timeseries. This was the favourable choice as opposed to min-max normalisation, since many detrital photosynthesis data already come expressed as % or proportion of initial. The drawback is that data aren't fully brought onto the same scale because some variables allow negative proportions (e.g. net gas exchange) while most don't (e.g. Chl, F_v/F_m etc.). I assume that the effect of this on the outcome of the analysis is minimal due to the relative scarcity of net gas exchange data and enforcing $\tau = 0$ (Equation 2).

Analysis was carried out as described above to model detrital viability over time, using a simplified Equation 2 with $a = 1$ and $\tau = 0$. Additive categorical predictors caused convergence issues, so I created a composite grouping variable from Group, Light and Response. Water is only relevant to a relatively small subset of terrestrial plant data and was therefore excluded as a predictor. No partial pooling was applied here because I did not want to make predictions for plants at large and introducing hyperparameters led to convergence issues. The gamma prior for k was centred on 0.22 d^{-1}, the mean across seagrasses which emerged from the experimental component of this study (Table 1) and is assumed to be intermediate between seaweeds and terrestrial plants. Experimental durations varied substantially across studies (Table S1), so half of the mean experimental duration was picked as the prior mean for μ. I again chose a reasonable s.d. based on prior simulation and put an additional ~~joint~~ prior on the product of k and μ. Several terrestrial plant studies reported photosynthetic decay over periods of only minutes post-excision (Table S1). These data pull the intercept down when given a weak joint prior, necessitating a much tighter prior. The logistic intercept was therefore constrained ~~to 4 ± 0.1, equivalent to 0.98 ± 0.0018 on the probability scale:~~with `target += normal`~~`gamma`~~`_lpdf( log(k) +`~~`.*`~~` log(mu) |`~~`–`~~` log(5) `~~`4^2 / 0.1^2`~~`, `~~`4 / 0.1^2`~~`0.05 ).`

Results

Experiment (the research)

Halophila ovalis photosynthesised for longer post-excision than *Amphibolis antarctica*. After 34 d, the duration of the experiment, all *H. ovalis* leaves were still obviously net autotrophic while all *A. antarctica* leaves were apparently dead. Comparing logistic decay rates (k) and times of death (μ) of light-saturated net photosynthesis (P_{max}) clarifies this difference. On a mass basis, *A. antarctica* started with ~~a similar~~somewhat higher P_{max} ~~to~~ than *H. ovalis*, but its P_{max} decayed 1.7~~5~~ times faster and expired 21~~0~~ ± 2.6~~4~~ d (mean ± s.d.) earlier (Figure 1a, Table 1). On a leaf basis, *A. antarctica* P_{max} only decayed ~~40%~~1.2 times faster and expired 14~~5~~ ± 2.9~~3~~ d earlier than that of *H. ovalis* (Figure 1b, Table 1). This difference between mass- and leaf-based estimates was primarily due to varying trends for *H. ovalis* (Figure 1a vs. b). While *A. antarctica* displayed comparable k (Δ = 0.0079~~0.038~~ ± 0.085~~0.059~~ d^{-1}, P = 0.51~~75~~) and μ (Δ = 0.26 ~~1.2~~ ± 1.9~~2~~ d, P = 0.55~~73~~), leaf-based *H. ovalis* P_{max} declined ~~53~~27% (P = 0.81~~92~~) faster and expired 7.3~~6.5~~ ± 3.5~~3~~ d (P = 0.98) earlier.

When accounting for leaf mass, *H. ovalis* thus seems to be able to photosynthesise longer post-excision. The immediate reason for this was the linear decline of leaf mass in *H. ovalis* (α = 0.052 ± 0.005 g leaf^{-1}, β = −0.58 ± 0.26 mg leaf^{-1} d^{-1} or −1.1 ± 0.43 % d^{-1}, $P_{\beta < 0}$ = 0.99) while *A. antarctica* leaf mass remained almost unchanged (α = 0.64 ± 0.053 g leaf^{-1}, β = −2 ± 2.7 mg leaf^{-1} d^{-1} or −0.3 ± 0.41 % d^{-1}, $P_{\beta<0}$ = 0.78). But it remains unclear why *H. ovalis* leaves of reduced mass photosynthesised disproportionally. The most likely reason is that ~~filamentous green algae~~periphyton, which I observed colonising *H. ovalis* but not *A. antarctica*, contributed much to photosynthesis but very little to leaf mass. This renders the leaf-based *H. ovalis* estimates more trustworthy and may have exacerbated the reported differences between the two seagrasses but is unlikely to have confounded the general trend.

Despite their phylogenetic and morphological diversity, both seagrasses displayed quite consistent tendencies, allowing relatively precise prediction for seagrasses in general (Figure 1, Table 1). The discussed discrepancy between mass- and leaf-based P_{max} of *H. ovalis* only marginally affected estimates on this higher level. Overall, seagrasses had similar k (Δ = 0.018~~0.0055~~ ± 0.24~~0.075~~ d^{-1}, P = 0.564) and μ (Δ = 2.4~~1.4~~ ± 12 ~~4.8~~ d, P = 0.58~~0.62~~) across mass- and leaf-based models (Table 1). It therefore seems that seagrass P_{max} generally decays at a rate of 0.21~~2~~ d^{-1} and persists for around ~~17~~ 21 d.

Meta-analysis (the context)

Comparison across plants revealed that only seaweeds can maintain detrital photosynthesis for several months. Seagrasses are limited to less than a month and freshwater and terrestrial plants to about a week (Figure 2, Table 2). Under light, k of seaweeds is 97~~6~~% (Δ = 0.79~~0.51~~ ± 0.057~~0.052~~ d^{-1}, P = 1) smaller than that of terrestrial plants. Seagrass photosynthesis decays 9.1~~7.5~~ times (Δ = 0.19~~0.15~~ ± 0.03~~0.025~~ d^{-1}, P = 1) faster than that of seaweeds and 74~~68~~% (Δ = 0.6~~0.36~~ ± 0.064~~0.058~~ d^{-1}, P = 1) slower than that of terrestrial plants. Correspondingly, μ of seaweeds occurs 273 ~~210~~ ± 129 ~~102~~ d (P = 1) later than in terrestrial plants. Seagrass detritus again takes an intermediate position, losing the ability to photosynthesise ~~190~~ 254 ± ~~102~~ 129 d (P = 1) earlier than seaweed detritus and 18 ± 3.5~~6~~ d (P = 1) later than terrestrial plant detritus.

Photosynthesis and chlorophyll (Chl) followed fairly consistent trends for seagrasses and terrestrial and freshwater plants (Figure 2, Table 2). This indicates that for these plants detrital Chl is a reasonable predictor of detrital photosynthesis. Seaweeds again form an exception and had 54~~69~~% smaller k and 1.1~~3~~ times larger μ for Chl than photosynthesis (Table 2), meaning that Chl tends to overestimate longevity of detrital photosynthesis ~~fourfold~~ twofold for this group (Figure 2). Nonetheless, since there are no data on photosynthetic senescence under light for freshwater plants, Chl enables a mostly representative comparison of all plant groups. Surprisingly, freshwater plant detritus lost Chl at similar rates to terrestrial plant detritus (Δ = 0.048~~0.023~~ ± 0.096~~0.087~~ d^{-1}, P = 0.7~~6~~), and consequently 1.64 times (Δ = 0.52 ~~0.36~~ ± 0.11~~0.094~~ d^{-1}, P = 1) and ~~102~~ 89 times (Δ = 0.84~~0.61~~ ± 0.072~~0.066~~ d^{-1}, P = 1) faster than seagrass and seaweed detritus respectively. Accordingly, photosynthetic death of freshwater plants practically occurs at the same time (Δ = 0.35~~0.21~~ ± 0.73~~0.91~~ d, P = 0.69) as terrestrial plants, and therefore 10 ± 4.34 d (P = 1) and ~~850~~ 566 ± ~~577~~ 159 d (P = 1) earlier than that of seagrasses and seaweeds respectively.

Light ameliorated photosynthetic decay in all except freshwater plants (Figure 2, Table 2). However, it only substantially extended detrital longevity in seaweeds. Whereas the effect is negligible for freshwater plants and only on the order of a couple days to a week for terrestrial plants and seagrasses, seaweed photosynthesis decays ~~25~~ 106 times faster and terminates 98~~6~~% earlier in darkness (Table 2). Consequently, under darkness seaweed detritus only remains viable for comparable durations to terrestrial (Δ = 0.91~~4.2~~ ± 9.3~~7.3~~ d, P = 0.54~~89~~) and freshwater (Δ = 1~~4.4~~ ± 9.3~~7.3~~ d, P = 0.54~~0.9~~) plants. This suggests that the extreme detrital longevity observed in seaweeds is highly light-dependent.

Discussion

Despite the existence of various relevant literature, the phenomenon of detrital photosynthesis had not been explored beyond kelps. In part, this is probably due to diverse research objectives which obscure the connection between quantitatively comparable data. There was also data scarcity in some important plant groups, especially the seagrasses. Here I filled this knowledge gap by presenting the first data on photosynthesis in seagrass detritus and subsequently the first meta-analysis of detrital photosynthesis across all plants.

Detrital photosynthesis *per se* is not restricted to non-vascular plants. Seagrass detritus can also photosynthesise for at least a couple weeks. The physiology of diverse seagrasses responds similarly to detrital age and observed differences may simply be attributable to varying colonisation by ~~biofilm algae~~periphyton which, given a leaf mass per area of 2.7 and 6.8 mg cm^{-2} for *Halophila ovalis* and *Amphibolis antarctica* respectively (de los Santos *et al.*, 2012, 2016), could contribute 12–172% of initial seagrass photosynthesis (Neely and Wetzel, 1997). Alternatively, ~~*Amphibolis*~~ *A. antarctica* leaves may be more reliant on their roots for nutrient provisioning. But there is no other evidence to suggest this (Pedersen *et al.*, 1997) and the 15–20 d earlier photosynthetic death of isolated *A. antarctica* leaves compared to *H.*~~*alophila*~~ *ovalis* is surprising given their 1.7 times (Δ = 47 d) longer attached lifespan (Hemminga *et al.*, 1999). Seagrass detrital photosynthesis, despite persisting for weeks, may not necessarily counteract decomposition, a process which can a year for these plants (Harrison, 1989; Cebrián *et al.*, 1997; Trevathan-Tackett *et al.*, 2020). While live *Halophila decipiens* ramets do indeed decompose slower than buried ones (Kenworthy *et al.*, 1989), decomposition rates are similar between live and senescent or buried *Thalassia testudinum* leaves (Rublee and Roman, 1982; Fourqurean and Schrlau, 2003; Rosch and Koch, 2009) and meta-analysis suggests that live seagrass leaves actually tend to decompose faster than senescent or dead ones (Harrison, 1989; Trevathan-Tackett *et al.*, 2020).

Macroalgae uniquely exhibit long-term detrital photosynthesis. In fact, there is nothing to suggest that detritus produced by these plants cannot remain viable for up to a year. As expected, seagrasses exhibit intermediate detrital longevity between seaweeds and terrestrial plants, which photosynthesise for at most one week post-excision. The extreme contrast to terrestrial plants has been demonstrated but is in fact somewhat dampened in my formal analysis. When given light but no water, which is the normal fate of terrestrial plant detritus, detached leaves of various trees tend to lose the ability to photosynthesise in just a few minutes (e.g. ~~(~~Gauthier and Jacobs, 2018; Kar *et al.*, 2021). This is due to breakdown of stomatal conductance following a loss in water pressure (Powles *et al.*, 2006; Gauthier and Jacobs, 2018). What surprised me is that detrital viability of freshwater plants is like that of terrestrial plants and not seagrasses. One explanation is excess production of hydrogen peroxide during photosynthesis in the detached leaf which leads to chlorophyll degradation, as demonstrated for *Hydrilla verticillata* (Kar and Choudhuri, 1986, 1987) and is likely also the case for other freshwater angiosperms (Jana and Choudhuri, 1982). In contrast, there is no evidence for increased chlorophyll breakdown in isolated seagrass leaves exposed to light (Strother and Vatta, 1986) and photodegradation is only apparent in dead leaves (Vähätalo *et al.*, 1998). Emancipation of leaf photosynthesis from the rest of the plant is apparently not favoured by an aquatic but rather by a marine nature in vascular plants. What exactly happened to allow this during the transition of angiosperms from lakes and rivers to the sea remains unclear. Perhaps the answer lies in convergent evolution with seaweeds, in particular alterations to the light harvesting complexes and cell wall features that enhance gas exchange (Olsen *et al.*, 2016)?

Senescence mechanisms determine the impact physiology can have on detrital dynamics. Seagrass detritus often incudes green leaves (Knauer and Ayers, 1977; Ochieng and Erftemeijer, 1999; Jiménez-Ramos *et al.*, 2023). This may be in part because nutrient resorption prior to abscission is low relative to terrestrial plants (Harrison, 1989; Pedersen and Borum, 1992; Pedersen *et al.*, 1997; Hemminga *et al.*, 1999; Rosch and Koch, 2009). But the most likely mechanism is premature detachment caused by external factors such as hydrodynamics and herbivory (Cebrián *et al.*, 1997) paired with seasonal weakening of the ligule (Jiménez-Ramos *et al.*, 2023). Viable seagrass detritus is clearly produced under natural conditions, but due to the mentioned mismatch between rapid photosynthetic decay and slow decomposition this is expected to have little influence on detrital dynamics. Seaweeds tell a different story. Their detritus can photosynthesise for many months, matching their decomposition timescale, if we assume that excision mirrors detrital export in nature. Distally eroded and abscised tissue almost certainly exhibits reduced detrital photosynthesis due to fragmentation and attached senescence, which can greatly accelerate decomposition (de Bettignies *et al.*, 2020). But up to half of kelp detritus is released by dislodgement of whole plants or large fragments (de Bettignies *et al.*, 2013; Pessarrodona *et al.*, 2018; Pedersen *et al.*, 2020). The visible detrital pool therefore consists mostly of tissue that displays limited senescence and probably can remain photosynthetic for many months.

I have previously suggested that detrital photosynthesis influences kelp blue carbon by providing an additional dimension to detrital recalcitrance (Wright *et al.*, 2022, 2024). Its uniqueness among plants, as evidenced here, suggests that it may play a disproportionate role in determining the fate of macroalgal carbon. The key implication is that seaweed decomposition rates are not directly comparable to those of other plants. It is even hard to make comparisons within seaweeds when not explicitly modelling detrital photosynthesis (Kennedy and Blain, 2025). Most of our understanding of seaweed decomposition is based on freshly excised blades (e.g. ~~(~~Albright *et al.*, 1982; Bedford and Moore, 1984; Frontier *et al.*, 2021; Filbee-Dexter *et al.*, 2022; Wright *et al.*, 2022) as opposed to senescent (de Bettignies *et al.*, 2020) or dead (Birch *et al.*, 1983; Smith and Foreman, 1984; Brouwer, 1996) fragments. But as mentioned, less than half of seaweed detritus can be considered fully viable when exported. Additionally, decomposition experiments are usually carried out in the shallow subtidal with access to light, but proponents of seaweed carbon sinks invariably place them in the darkness of the deep sea (Krause-Jensen and Duarte, 2016; Filbee-Dexter *et al.*, 2024). My meta-analysis suggests that darkness has a stronger detrimental effect on detrital photosynthesis in seaweeds than any other plant. If detrital photosynthesis is granted the ecological significance I attribute to it, these contradictions must make one wonder. Perhaps it is time to question existing preconceptions in seaweed blue carbon?

To conclude, I show that vascular plants also photosynthesise after detachment, but this varies greatly from a few minutes in tree leaves without access to water, over several days on average in terrestrial and freshwater plants to a couple of weeks in seagrasses. Macroalgae alone can photosynthesise for months post-excision. The powerful effect of detrital photosynthesis on decomposition and thereby blue carbon has already been evidenced to some extent (Birch *et al.*, 1983; Brouwer, 1996; de Bettignies *et al.*, 2020). What now remains to be investigated is (1) decomposition trajectories of macroalgal detritus with and without physiology and how these compare with seagrasses as the most functionally similar plants, (2) an improved decomposition model for macroalgae that explicitly accounts for detrital photosynthesis, (~~2~~3) the prevalence of photosynthesis in eroding, abscised and other senescing tissues, and (~~3~~4) the proportion of the detrital pool that photosynthesises *in situ*.

Supplementary data

Supplementary data are available at *Annals of Botany* online and consist of the following.

Figure S1: Effect of confounding variables associated with incubation on light-saturated net seagrass photosynthesis. Table S1: Species included in meta-analysis. References: Studies included in the meta-analysis in addition to the present study.

Funding

This work was supported by the Forrest Research Foundation Scholarship and Australian Government Research Training Program Scholarship. Open access publishing was facilitated by The University of Western Australia, as part of the Oxford University Press–The University of Western Australia agreement via the Council of Australian University Librarians.

Acknowledgements

Thank you to Andy Foggo and Nadia Frontier for inspiring discussions, Belinda Martin for introducing me to *Halophila ovalis* and joining me in the field, Thomas Wernberg for supplying and Chunbo Liu for repairing the oxygen meter, ~~Taylor Simpkins and Mason Sullivan for technical support,~~ and Nadia Frontier and Dan van Hees for sharing data.

Data availability

Experimental data and annotated code are available at github.com/lukaseamus/seagrass-detrital-photosynthesis. The dataset used for meta-analysis is published (Wright, 2025) and can also be accessed at github.com/lukaseamus/detrital-photosynthesis. I place no restrictions on data and code availability within the constraints of the specified copyleft licence: GNU General Public License.

Figure legends

Figure 1. Seagrass detrital photosynthesis. Light-saturated net photosynthesis (P_{max}) per gram of blotted mass (**A**) and per leaf (*H. ovalis*) or leaf cluster (*A. antarctica*) (**B**) as a function of time post-excision. Distributions are kernel density estimates of priors and posteriors for parameters (Equation 2) ~~with vertical lines demarking the central 50, 80 and 90% of probability density~~. In **B**, distributions for α and τ are not shown because they were modelled on the sample mass scale with an appropriate prior that cannot be converted to the leaf mass scale (see Table 1 for numerical estimates of α and τ). Violins are kernel density estimates of posterior probability distributions for observations (Equation 1), showing the measurement error which is associated with my method and was incorporated into the model. Lines and intervals are medians and the

central 50, 80 and 90% of posterior probability for P_{μ}, the mean prediction of P_{max} (Equation 2). Light grey lines encompass the central 90% of prior probability.

Figure 2. Detrital photosynthesis meta-analysis. Photosynthesis (*P*) and chlorophyll (*Chl*) variables as a function of time post-excision across four major plant groups: terrestrial plants (Streptophyta excluding aquatic plants), freshwater plants (Alismatales excluding seagrasses), seagrasses, and seaweeds (Chlorophyta and Heterokontophyta excluding freshwater macroalgae). Distributions are kernel density estimates of priors and posteriors for parameters~~ with vertical lines demarking the central 50, 80 and 90% of probability density~~. Points are observations and lines and intervals are medians and the central 50, 80 and 90% of posterior probability for the mean prediction. Light grey lines encompass the central 90% of prior probability. Note that several observations outside prior probability space were excluded from the plot for clarity but not from the analysis.

Literature cited

Albright LJ, Chocair J, Masuda K, Valdes M. **1982**. Degradation of the kelps *Macrocystis integrifolia* and *Nereocystis luetkeana* in British Columbia coastal waters. In Srivastava LM (ed) Synthetic and Degradative Processes in Marine Macrophytes. De Gruyter, Berlin, pp 215-234. https://doi.org/10.1515/9783110837988-015.

Apostolaki ET, Lavery PS, Litsi-Mizan V *et al.* **2024**. Patterns of carbon and nitrogen accumulation in seagrass (*Posidonia oceanica*) meadows of the Eastern Mediterranean Sea. *Journal of Geophysical Research: Biogeosciences* **129**: e2024JG008163. https://doi.org/10.1029/2024jg008163.

Bedford AP, Moore PG. **1984**. Macrofaunal involvement in the sublittoral decay of kelp debris: the detritivore community and species interactions. *Estuarine, Coastal and Shelf Science* **18**: 97–111. https://doi.org/10.1016/0272-7714(84)90009-x.

de Bettignies F, Dauby P, Thomas F *et al.* **2020**. Degradation dynamics and processes associated with the accumulation of *Laminaria hyperborea* (Phaeophyceae) kelp fragments: an in situ experimental approach. *Journal of Phycology* **56**: 1481–1492. https://doi.org/10.1111/jpy.13041.

de Bettignies T, Wernberg T, Lavery PS, Vanderklift MA, Mohring MB. **2013**. Contrasting mechanisms of dislodgement and erosion contribute to production of kelp detritus. *Limnology and Oceanography* **58**: 1680–1688. https://doi.org/10.4319/lo.2013.58.5.1680.

Birch PB, Gabrielson JO, Hamel KS. **1983**. Decomposition of *Cladophora*. I. Field studies in the Peel-Harvey estuarine system, Western Australia. *Botanica Marina* **26**: 165–172. https://doi.org/10.1515/botm.1983.26.4.165.

Björk M, Weil A, Semesi S, Beer S. **1997**. Photosynthetic utilisation of inorganic carbon by seagrasses from Zanzibar, East Africa. *Marine Biology* **129**: 363–366. https://doi.org/10.1007/s002270050176.

Blum LK, Mills AL. **1991**. Microbial growth and activity during the initial stages of seagrass decomposition. *Marine Ecology Progress Series* **70**: 73–82. https://doi.org/10.3354/meps070073.

Bolton JJ. **2016**. What is aquatic botany? — and why algae are plants: the importance of non-taxonomic terms for groups of organisms. *Aquatic Botany* **132**: 1–4. https://doi.org/10.1016/j.aquabot.2016.02.006.

Borum J, Pedersen O, Kotula L *et al.* **2016**. Photosynthetic response to globally increasing CO_2 of co-occurring temperate seagrass species. *Plant, Cell & Environment* **39**: 1240–1250. https://doi.org/10.1111/pce.12658.

Brouwer PEM. **1996**. Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology* **16**: 129–137. https://doi.org/10.1007/bf02390433.

Carpenter B, Gelman A, Hoffman MD *et al.* **2017**. Stan: a probabilistic programming language. *Journal of Statistical Software* **76**: 1–32. https://doi.org/10.18637/jss.v076.i01.

Cebrián J, Duarte CM, Marbà N, Enríquez S. **1997**. Magnitude and fate of the production of four co-occurring Western Mediterranean seagrass species. *Marine Ecology Progress Series* **155**: 29–44. https://doi.org/10.3354/meps155029.

Evans AS, Webb KL, Penhale PA. **1986**. Photosynthetic temperature acclimation in two coexisting seagrasses, *Zostera marina* L. and *Ruppia maritima* L. *Aquatic Botany* **24**: 185–197. https://doi.org/10.1016/0304-3770(86)90095-1.

Evrard V, Kiswara W, Bouma TJ, Middelburg JJ. **2005**. Nutrient dynamics of seagrass ecosystems: ^{15}N evidence for the importance of particulate organic matter and root systems. *Marine Ecology Progress Series* **295**: 49–55. https://doi.org/10.3354/meps295049.

Filbee-Dexter K, Feehan CJ, Smale DA *et al.* **2022**. Kelp carbon sink potential decreases with warming due to accelerating decomposition. *PLoS Biology* **20**: e3001702. https://doi.org/10.1371/journal.pbio.3001702.

Filbee-Dexter K, Pessarrodona A, Pedersen MF *et al.* **2024**. Carbon export from seaweed forests to deep ocean sinks. *Nature Geoscience* **17**: 552–559. https://doi.org/10.1038/s41561-024-01449-7.

Fourqurean JW, Schrlau JE. **2003**. Changes in nutrient content and stable isotope ratios of C and N during decomposition of seagrasses and mangrove leaves along a nutrient availability gradient in Florida Bay, USA. *Chemistry and Ecology* **19**: 373–390. https://doi.org/10.1080/02757540310001609370.

Fraser CI, Morrison AK, Hogg AM *et al.* **2018**. Antarctica's ecological isolation will be broken by storm-driven dispersal and warming. *Nature Climate Change* **8**: 704–708. https://doi.org/10.1038/s41558-018-0209-7.

Frontier N, de Bettignies F, Foggo A, Davoult D. **2021**. Sustained productivity and respiration of degrading kelp detritus in the shallow benthos: detached or broken, but not dead. *Marine Environmental Research* **166**: 105277. https://doi.org/10.1016/j.marenvres.2021.105277.

Gabry J, Češnovar R, Johnson A, Bronder S. **2024**. cmdstanr: R Interface to 'CmdStan'. R package version 0.5.3. https://mc-stan.org/cmdstanr.

Gabry J, Simpson D, Vehtari A, Betancourt M, Gelman A. **2019**. Visualization in Bayesian workflow. *Journal of the Royal Statistical Society Series A: Statistics in Society* **182**: 389–402. https://doi.org/10.1111/rssa.12378.

Gauthier M-M, Jacobs DF. **2018**. Reductions in net photosynthesis and stomatal conductance vary with time since leaf detachment in three deciduous angiosperms. *Trees* **32**: 1247–1252. https://doi.org/10.1007/s00468-018-1706-z.

Graiff A, Karsten U, Meyer S, Pfender D, Tala F, Thiel M. **2013**. Seasonal variation in floating persistence of detached *Durvillaea antarctica* (Chamisso) Hariot thalli. *Botanica Marina* **56**: 3–14. https://doi.org/10.1515/bot-2012-0193.

Graiff A, Pantoja JF, Tala F, Thiel M. **2016**. Epibiont load causes sinking of viable kelp rafts: seasonal variation in floating persistence of giant kelp *Macrocystis pyrifera*. *Marine Biology* **163**: 191. https://doi.org/10.1007/s00227-016-2962-3.

Gras AF, Koch MS, Madden CJ. **2003**. Phosphorus uptake kinetics of a dominant tropical seagrass *Thalassia testudinum*. *Aquatic Botany* **76**: 299–315. https://doi.org/10.1016/s0304-3770(03)00069-x.

Guiry, M.D., Guiry GM. **2025**. *AlgaeBase*. https://www.algaebase.org. 7th February 2025

Harrison PG. **1989**. Detrital processing in seagrass systems: a review of factors affecting decay rates, remineralization and detritivory. *Aquatic Botany* **35**: 263–288. https://doi.org/10.1016/0304-3770(89)90002-8.

Hemminga MA, Marba N, Stapel J. **1999**. Leaf nutrient resorption, leaf lifespan and the retention of nutrients in seagrass systems. *Aquatic Botany* **65**: 141–158. https://doi.org/10.1016/s0304-3770(99)00037-6.

Jamaludin MR, Bujang JS, Kusnan M, Omar HO. **2006**. Photosynthetic light responses of wild and cultured *Halophila ovalis*. *Malaysian Journal of Science* **25**: 67–79.

Jana S, Choudhuri MA. **1980**. Senescence in submerged aquatic angiosperms: changes in intact and isolated leaves during aging. *New Phytologist* **86**: 191–198. https://doi.org/10.1111/j.1469-8137.1980.tb03188.x.

Jana S, Choudhuri MA. **1982**. Glycolate metabolism of three submersed aquatic angiosperms during ageing. *Aquatic Botany* **12**: 345–354. https://doi.org/10.1016/0304-3770(82)90026-2.

Jiménez-Ramos R, Henares C, Egea LG, Vergara JJ, Brun FG. **2023**. Leaf senescence of the seagrass *Cymodocea nodosa* in Cádiz Bay, Southern Spain. *Diversity* **15**: 187. https://doi.org/10.3390/d15020187.

Kar RK, Choudhuri MA. **1987**. Possible mechanisms of light-induced chlorophyll degradation in senescing leaves of *Hydrilla verticillata*. *Physiologia Plantarum* **70**: 729–734. https://doi.org/10.1111/j.1399-3054.1987.tb04331.x.

Kar RK, Choudhuri MA. **1986**. Effects of light and spermine on senescence of *Hydrilla* and spinach leaves. *Plant Physiology* **80**: 1030–1033. https://doi.org/10.1104/pp.80.4.1030.

Kar S, Montague DT, Villanueva-Morales A. **2021**. Measurement of photosynthesis in excised leaves of ornamental trees: a novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees* **35**: 889–905. https://doi.org/10.1007/s00468-021-02088-w.

Kay M. **2024**. tidybayes: tidy data and geoms for Bayesian models v3.0.7. *Zenodo*. https://doi.org/10.5281/zenodo.13770114.

Kennedy JR, Blain CO. **2025**. A systematic review of marine macroalgal degradation: toward a better understanding of macroalgal carbon sequestration potential. *Journal of Phycology* https://doi.org/10.1111/jpy.70031.

Kenworthy WJ, Currin CA, Fonseca MS, Smith G. **1989**. Production, decomposition, and heterotrophic utilization of the seagrass *Halophila decipiens* in a submarine canyon. *Marine Ecology Progress Series* **51**: 277–290. https://doi.org/10.3354/meps051277.

Knauer GA, Ayers AV. **1977**. Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography* **22**: 408–414. https://doi.org/10.4319/lo.1977.22.3.0408.

Krause-Jensen D, Duarte CM. **2016**. Substantial role of macroalgae in marine carbon sequestration. *Nature Geoscience* **9**: 737–742. https://doi.org/10.1038/ngeo2790.

Lamit N, Tanaka Y. **2021**. Effects of river water inflow on the growth, photosynthesis, and respiration of the tropical seagrass *Halophila ovalis*. *Botanica Marina* **64**: 93–100. https://doi.org/10.1515/bot-2020-0079.

Lee D, Stan buildbot, seantalts *et al.* **2017**. stan-dev/cmdstan: v2.17.1. *Zenodo*. https://doi.org/10.5281/zenodo.1117248.

Lee K-S, Dunton KH. **1999**. Inorganic nitrogen acquisition in the seagrass *Thalassia testudinum*: development of a whole-plant nitrogen budget. *Limnology and Oceanography* **44**: 1204–1215. https://doi.org/10.4319/lo.1999.44.5.1204.

Lovell PH, Illsley ANN, Moore KG. **1972**. The effects of light intensity and sucrose on root formation, photosynthetic ability, and senescence in detached cotyledons of *Sinapis alba* L. and *Raphanus sativus* L. *Annals of Botany* **36**: 123–134. https://doi.org/10.1093/oxfordjournals.aob.a084565.

Macaya EC, Boltana S, Hinojosa IA *et al.* **2005**. Presence of sporophylls in floating kelp rafts of *Macrocystis* spp. (Phaeophyceae) along the Chilean Pacific coast. *Journal of Phycology* **41**: 913–922. https://doi.org/10.1111/j.1529-8817.2005.00118.x.

Marbà N, Hemminga MA, Mateo MA *et al.* **2002**. Carbon and nitrogen translocation between seagrass ramets. *Marine Ecology Progress Series* **226**: 287–300. https://doi.org/10.3354/meps226287.

Masini RJ, Manning CR. **1997**. The photosynthetic responses to irradiance and temperature of four meadow-forming seagrasses. *Aquatic Botany* **58**: 21–36. https://doi.org/10.1016/s0304-3770(97)00008-9.

Mateo MA, Romero J. **1996**. Evaluating seagrass leaf litter decomposition: an experimental comparison between litter-bag and oxygen-uptake methods. *Journal of Experimental Marine Biology and Ecology* **202**: 97–106. https://doi.org/10.1016/0022-0981(96)00019-6.

Mateo MA, Romero J. **1997**. Detritus dynamics in the seagrass *Posidonia oceanica*: elements for an ecosystem carbon and nutrient budget. *Marine Ecology Progress Series* **151**: 43–53. https://doi.org/10.3354/meps151043.

Mohr W, Lehnen N, Ahmerkamp S *et al.* **2021**. Terrestrial-type nitrogen-fixing symbiosis between seagrass and a marine bacterium. *Nature* **600**: 105–109. https://doi.org/10.1038/s41586-021-04063-4.

Moore JC, Berlow EL, Coleman DC *et al.* **2004**. Detritus, trophic dynamics and biodiversity. *Ecology Letters* **7**: 584–600. https://doi.org/10.1111/j.1461-0248.2004.00606.x.

Neely RK, Wetzel RG. **1997**. Autumnal production by bacteria and autotrophs attached to *Typha latifolia* L. detritus. *Journal of Freshwater Ecology* **12**: 253–267. https://doi.org/10.1080/02705060.1997.9663533.

Ochieng CA, Erftemeijer PLA. **1999**. Accumulation of seagrass beach cast along the Kenyan coast: a quantitative assessment. *Aquatic Botany* **65**: 221–238. https://doi.org/10.1016/s0304-3770(99)00042-x.

Olsen JL, Rouzé P, Verhelst B *et al.* **2016**. The genome of the seagrass *Zostera marina* reveals angiosperm adaptation to the sea. *Nature* **530**: 331–335. https://doi.org/10.1038/nature16548.

Patriquin D, Knowles R. **1972**. Nitrogen fixation in the rhizosphere of marine angiosperms. *Marine Biology* **16**: 49–58. https://doi.org/10.1007/bf00347847.

Pedersen MF, Filbee-Dexter K, Norderhaug KM *et al.* **2020**. Detrital carbon production and export in high latitude kelp forests. *Oecologia* **192**: 227–239. https://doi.org/10.1007/s00442-019-04573-z.

Pedersen MF, Paling EI, Walker DI. **1997**. Nitrogen uptake and allocation in the seagrass *Amphibolis antarctica*. *Aquatic Botany* **56**: 105–117. https://doi.org/10.1016/s0304-3770(96)01100-x.

Pedersen MF, Borum J. **1992**. Nitrogen dynamics of eelgrass *Zostera marina* during low nutrient availability. *Marine Ecology Progress Series* **80**: 65–73. https://doi.org/10.3354/meps080065.

Pellikaan GC. **1982**. Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological Bulletin* **16**: 83–92. https://doi.org/10.1007/bf02255416.

Pessarrodona A, Moore PJ, Sayer MDJ, Smale DA. **2018**. Carbon assimilation and transfer through kelp forests in the NE Atlantic is diminished under a warmer ocean climate. *Global Change Biology* **24**: 4386–4398. https://doi.org/10.1111/gcb.14303.

Powles JE, Buckley TN, Nicotra AB, Farquhar GD. **2006**. Dynamics of stomatal water relations following leaf excision. *Plant, Cell & Environment* **29**: 981–992. https://doi.org/10.1111/j.1365-3040.2005.01491.x.

POWO. **2025**. *Plants of the World Online. Facilitated by the Royal Botanic Gardens, Kew.* https://powo.science.kew.org. 7th February 2025

R Core Team. **2025**. *R: a language and environment for statistical computing*. Vienna: R Foundation for Statistical Computing.

Rohatgi A. **2025**. *WebPlotDigitizer v5.2*. https://automeris.io.

Rosch KL, Koch MS. **2009**. Nitrogen and phosphorus recycling by a dominant tropical seagrass (*Thalassia testudinum*) across a nutrient gradient in Florida Bay. *Bulletin of Marine Science* **84**: 1–24.

RStudio Team. **2025**. *RStudio: integrated development environment for R*. Boston: RStudio, Inc.

Rublee PA, Roman MR. **1982**. Decomposition of turtlegrass (*Thalassia testudinum* Konig) in flowing sea-water tanks and litterbags: compositional changes and comparison with natural particulate matter. *Journal of Experimental Marine Biology and Ecology* **58**: 47–58. https://doi.org/10.1016/0022-0981(82)90096-x.

Said N, Webster C, Dunham N, Strydom S, McMahon K. **2024**. Seagrass thermal tolerance varies between species and within species across locations. Prepared for the WAMSI Westport Marine Science Program. Western Australian Marine Science Institution, Perth, Western Australia. https://wamsi.org.au/wp-content/uploads/2024/10.

Said NE, McMahon K, Lavery PS. **2021**. Accounting for the influence of temperature and location when predicting seagrass (*Halophila ovalis*) photosynthetic performance. *Estuarine, Coastal and Shelf Science* **257**: 107414. https://doi.org/10.1016/j.ecss.2021.107414.

de los Santos CB, Onoda Y, Vergara JJ *et al.* **2016**. A comprehensive analysis of mechanical and morphological traits in temperate and tropical seagrass species. *Marine Ecology Progress Series* **551**: 81–94. https://doi.org/10.3354/meps11717.

de los Santos CB, Brun FG, Onoda Y *et al.* **2012**. Leaf-fracture properties correlated with nutritional traits in nine Australian seagrass species: implications for susceptibility to herbivory. *Marine Ecology Progress Series* **458**: 89–102. https://doi.org/10.3354/meps09757.

Short FT, McRoy CP. **1984**. Nitrogen uptake by leaves and roots of the seagrass *Zostera marina* L. *Botanica Marina* **27**: 547–555. https://doi.org/10.1515/botm.1984.27.12.547.

Smith BD, Foreman RE. **1984**. An assessment of seaweed decomposition within a southern Strait of Georgia seaweed community. *Marine Biology* **84**: 197–205. https://doi.org/10.1007/bf00393005.

Stapel J, Aarts TL, van Duynhoven BHM, de Groot JD, van den Hoogen PHW, Hemminga MA. **1996**. Nutrient uptake by leaves and roots of the seagrass *Thalassia hemprichii* in the Spermonde Archipelago, Indonesia. *Marine Ecology Progress Series* **134**: 195–206. https://doi.org/10.3354/meps134195.

Strother S, Vatta R. **1986**. The effects of light, cycloheximide and incubation medium on the senescence of detached seagrass leaves. *Biologia Plantarum* **28**: 211–218. https://doi.org/10.1007/bf02894599.

Tala F, López BA, Velásquez M *et al.* **2019**. Long-term persistence of the floating bull kelp *Durvillaea antarctica* from the South-East Pacific: potential contribution to local and transoceanic connectivity. *Marine Environmental Research* **149**: 67–79. https://doi.org/10.1016/j.marenvres.2019.05.013.

Thimann KV, Satler SO. **1979**. Relation between leaf senescence and stomatal closure: senescence in light. *Proceedings of the National Academy of Sciences* **76**: 2295–2298. https://doi.org/10.1073/pnas.76.5.2295.

Trevathan-Tackett SM, Jeffries TC, Macreadie PI, Manojlovic B, Ralph P. **2020**. Long-term decomposition captures key steps in microbial breakdown of seagrass litter. *Science of The Total Environment* **705**: 135806. https://doi.org/10.1016/j.scitotenv.2019.135806.

Vähätalo A, Søndergaard M, Schlüter L, Markager S. **1998**. Impact of solar radiation on the decomposition of detrital leaves of eelgrass *Zostera marina*. *Marine Ecology Progress Series* **170**: 107–117. https://doi.org/10.3354/meps170107.

Viana IG, Saavedra-Hortúa DA, Mtolera M, Teichberg M. **2019**. Different strategies of nitrogen acquisition in two tropical seagrasses under nitrogen enrichment. *New Phytologist* **223**: 1217–1229. https://doi.org/10.1111/nph.15885.

Waycott M, Biffin E, Les DH. **2018**. Systematics and evolution of Australian seagrasses in a global context. In Larkum AWD, Kendrick GA, Ralph PJ (eds) Seagrasses of Australia: Structure, Ecology and Conservation. Springer, Cham, pp 129-154. https://doi.org/10.1007/978-3-319-71354-0_5.

Wickham H, Averick M, Bryan J *et al.* **2019**. Welcome to the Tidyverse. *Journal of Open Source Software* **4**: 1686. https://doi.org/10.21105/joss.01686.

Wright LS, Foggo A. **2021**. Photosynthetic pigments of co-occurring Northeast Atlantic *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series* **678**: 227–232. https://doi.org/10.3354/meps13886.

Wright LS, Pessarrodona A, Foggo A. **2022**. Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. *Global Change Biology* **28**: 5514–5531. https://doi.org/10.1111/gcb.16299.

Wright LS, Kregting L. **2023**. Genus-specific response of kelp photosynthetic pigments to decomposition. *Marine Biology* **170**: 144. https://doi.org/10.1007/s00227-023-04289-y.

Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T. **2024**. Temperature sensitivity of detrital photosynthesis. *Annals of Botany* **133**: 17–28. https://doi.org/10.1093/aob/mcad167.

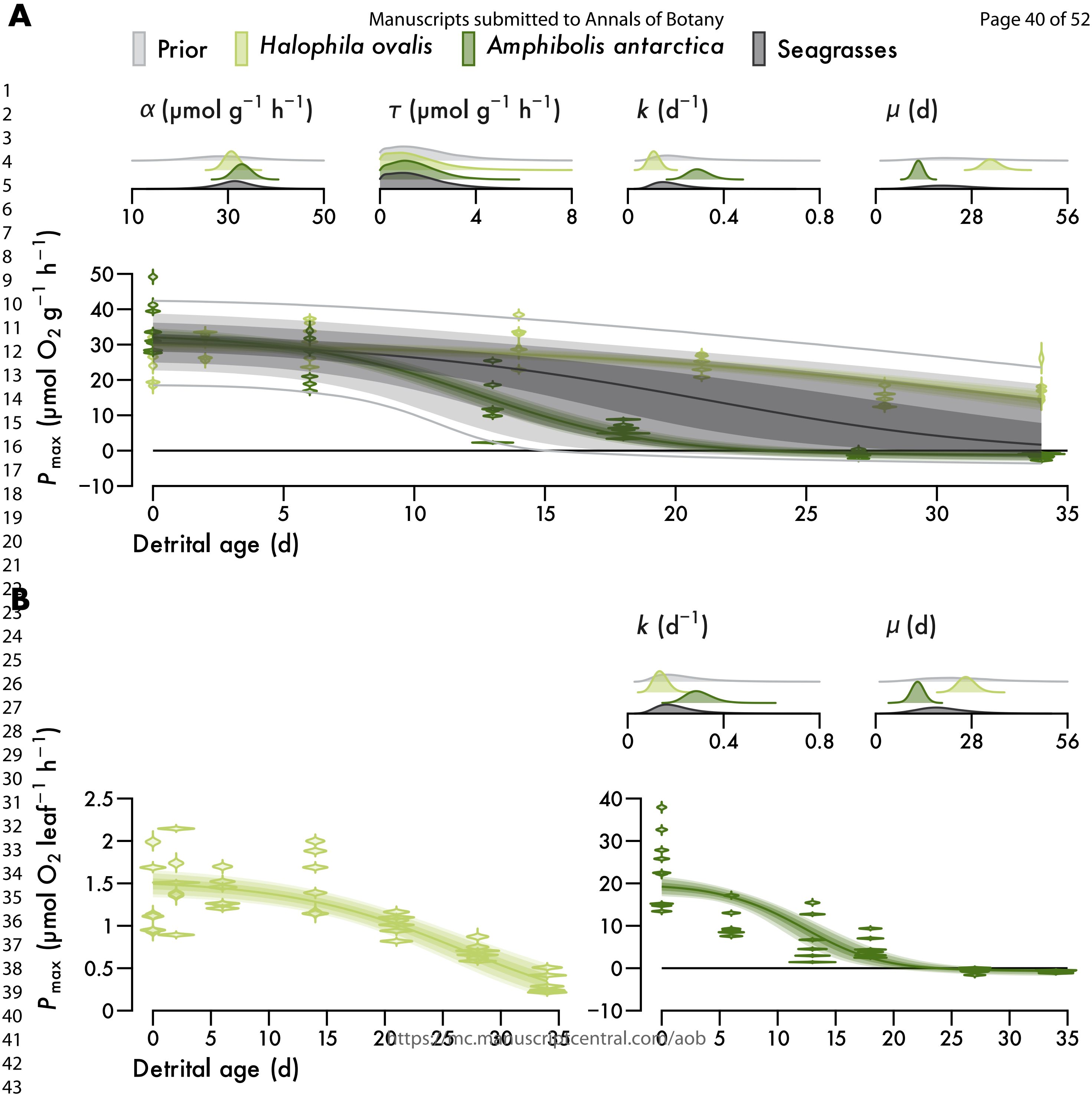
Wright LS. **2025**. Detrital photosynthesis (Dataset). *Zenodo*. https://doi.org/10.5281/zenodo.15054888.

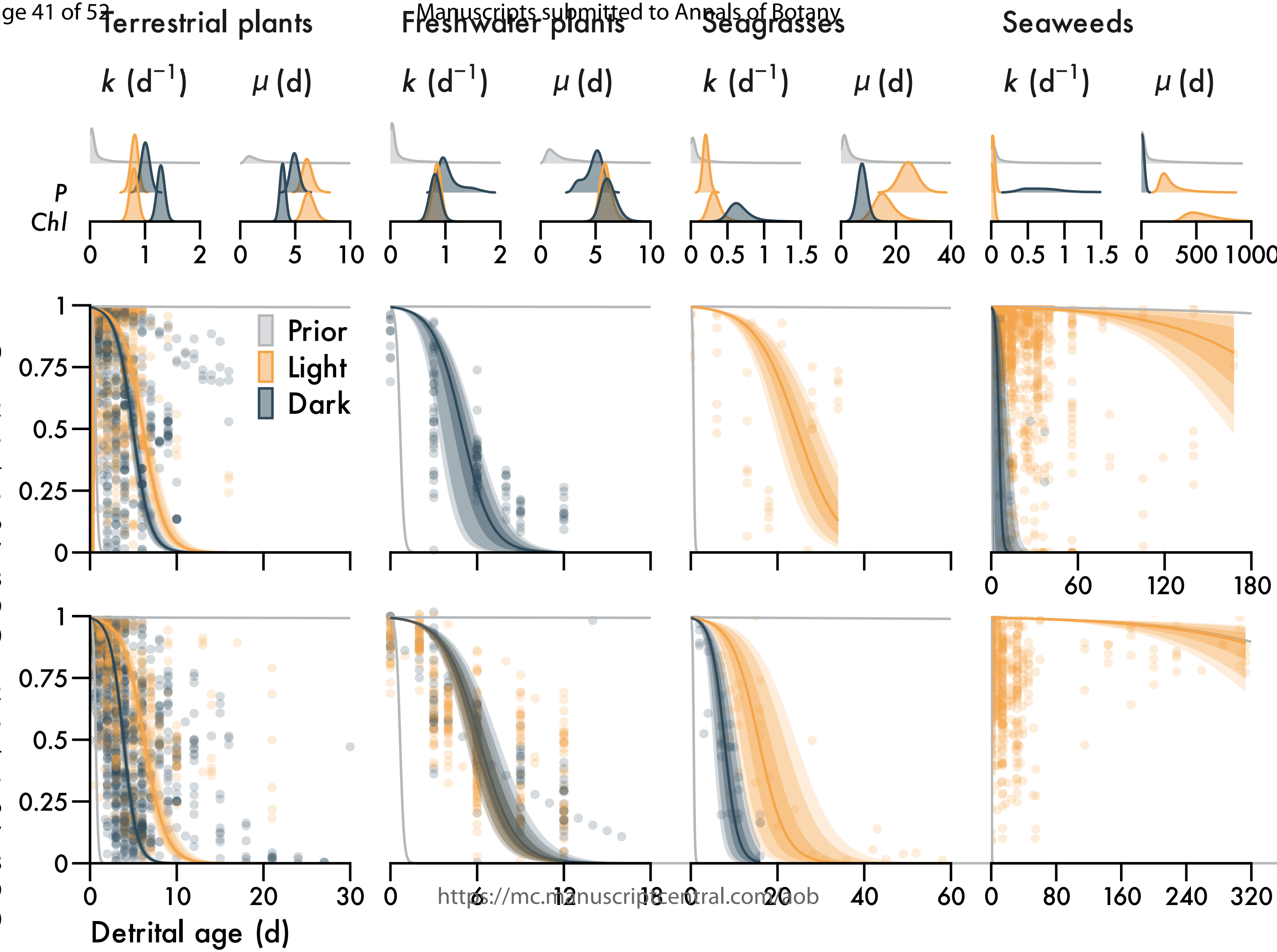
Table 1. General and specific seagrass parameter estimates (mean ± s.d., Equation 2) based on n = 39 (*Amphibolis antarctica*) to 42 (*Halophila ovalis*) leaf samples. 8×10^4 ~~8^4~~ posterior samples were drawn from the parameter distributions to calculate these estimates. Leaf-based α and τ were not estimated ~~across~~ for unobserved seagrasses since generalisation on that scale is not insightful and models were already run on a sample mass basis. Δ is the absolute difference between estimates for *H.* ~~*alophila*~~ *ovalis* and *A.* ~~*mphibolis*~~ *antarctica*. P is the probability that estimates for *H.* ~~*alophila*~~ *ovalis* and *A.* ~~*mphibolis*~~ *antarctica* are different, given the priors and data. P = 0.5 is equivalent to a coin toss. P > 0.9 is highlighted in bold.

	Seagrasses	*Halophila ovalis*	*Amphibolis antarctica*	Δ	P
Mass-based					
α (µmol g^{-1} h^{-1})	32 ± 5.2~~1.7~~	31~~32~~ ± 1.4~~1.6~~	33 ± 1.8	2.2 ~~1.1~~ ± 1.8	0.89~~0.74~~
τ (µmol g^{-1} h^{-1})	1.7~~1.4~~ ± 1.5~~0.93~~	1.5~~1.2~~ ± 1.3~~1.2~~	1.4~~1.3~~ ± 0.97~~1.1~~	0.12~~0.052~~ ± 1.3~~1.1~~	0.51~~0.54~~
k (d^{-1})	0.2~~2~~ ± 0.15~~0.055~~	0.11 ± 0.019~~0.023~~	0.3~~0.27~~ ± 0.044~~0.043~~	0.19~~0.16~~ ± 0.03~~0.047~~	**1**
μ (d)	22~~18~~ ± 9.1~~3.6~~	34~~32~~ ± 2.5~~2.3~~	12 ± 0.94~~1~~	21~~20~~ ± 2.6~~2.4~~	**1**
Leaf-based					
α (µmol leaf^{-1} h^{-1})		1.5 ± 0.11	20~~22~~ ± 1.6~~2~~	18~~20~~ ± 1.6~~2~~	**1**
τ (µmol leaf^{-1} h^{-1})		0.089~~0.079~~ ± 0.072~~0.063~~	0.79~~0.91~~ ± 0.59~~0.67~~	0.71~~0.83~~ ± 0.6~~0.67~~	**0.96~~7~~**
k (d^{-1})	0.22 ± 0.18~~0.051~~	0.14~~0.16~~ ± 0.027~~0.034~~	0.3~~0.23~~ ± 0.073~~0.041~~	0.17~~0.066~~ ± 0.06~~0.054~~	~~**0.91**~~ **1**
μ (d)	20~~16~~ ± 7.8~~3.1~~	27~~26~~ ± 2.5~~2.3~~	12~~11~~ ± 1.7	14~~15~~ ± 2.9~~3~~	**1**

Table 2. Plant group parameter estimates (mean ± s.d., Equation 2) based on n = 10566 observations and calculated with 8×10^4~~8~~^{4} posterior samples. Only k and μ are shown since response units were normalised and α and τ were therefore set to 1 and 0 respectively. Δ is the absolute difference between light and dark estimates within photosynthesis or chlorophyll. Δ with subscript is the absolute difference between photosynthesis and chlorophyll within light (Δ_L) or dark (Δ_D). P as in Table 1, i.e. the probability mass of Δ > 0. P = 0.5 is equivalent to a coin toss. P > 0.9 is highlighted in bold. Selected pairwise differences between plant groups are provided in text.

	Photosynthesis				**Chlorophyll**							
	Light	**Dark**	**Δ**	**P**	**Light**	**Dark**	**Δ**	**P**	**Δ_L**	**P_L**	**Δ_D**	**P_D**
Terrestrial plants												
k (d^{-1})	0.81~~0.53~~ ± 0.057~~0.052~~	1~~0.75~~ ± 0.069~~0.06~~	0.2~~2~~ ± 0.089~~0.079~~	**0.99~~1~~**	0.8~~0.59~~ ± 0.064~~0.055~~	1.3~~0.95~~ ± 0.059~~0.053~~	0.49~~0.36~~ ± 0.087~~0.077~~	**1**	0.013~~0.069~~ ± 0.085~~0.076~~	0.56~~0.82~~	0.28~~1~~ ± 0.091~~0.08~~	**1**
μ (d)	6.1~~6.7~~ ± 0.44~~0.64~~	5~~.2~~ ± 0.34~~0.39~~	1.2~~1.5~~ ± 0.55~~0.75~~	**0.99~~8~~**	6.3~~6.7~~ ± 0.51~~0.61~~	3.9~~4~~ ± 0.17~~0.19~~	2.4~~6~~ ± 0.54~~0.64~~	**1**	0.15~~0.011~~ ± 0.67~~0.88~~	0.59~~0.51~~	1.1~~2~~ ± 0.38~~0.44~~	**1**
Freshwater plants												
k (d^{-1})		1.1~~0.81~~ ± 0.21~~0.11~~			0.85~~0.62~~ ± 0.072~~0.066~~	0.81~~0.62~~ ± 0.091~~0.077~~	0.03~~0.0045~~ ± 0.12	0.6~~5~~1			0.26~~0.19~~ ± 0.23~~0.13~~	**0.92~~4~~**
μ (d)		4.8~~5~~ ± 0.8~~0.61~~			5.9~~6.5~~ ± 0.52~~0.67~~	6.2~~6.5~~ ± 0.7~~0.79~~	0.26~~0.032~~ ± 0.87~~1~~	0.62~~0.51~~			1.4~~5~~ ± 1.1	**0.92~~4~~**
Seagrasses												
k (d^{-1})	0.21~~0.17~~ ± 0.03~~0.025~~				0.33~~0.26~~ ± 0.087~~0.066~~	0.68~~0.51~~ ± 0.17~~0.12~~	0.35~~0.25~~ ± 0.19~~0.13~~	**0.98**	0.12~~0.093~~ ± 0.091~~0.071~~	**0.93~~2~~**		
μ (d)	25 ± 3.4~~3.6~~				16~~17~~ ± 4.3~~4.4~~	7.8~~8.2~~ ± 1.7~~1.9~~	8.4~~8.3~~ ± 4.6~~4.7~~	**0.98**	8.4~~1~~ ± 5.5~~6~~	**0.94~~3~~**		
Seaweeds												
k (d^{-1})	0.021~~0.02~~ ± 0.0064~~0.0047~~	2.2~~0.51~~ ± 2.6~~0.24~~	2.2~~0.49~~ ± 2.6~~0.24~~	**1**	0.0094~~0.006~~ ± 0.0024~~0.0025~~				0.011~~4~~ ± 0.0069~~0.0053~~	**0.94~~9~~**		
μ (d)	279~~218~~ ± 129~~102~~	5.9~~9.4~~ ± 9.3~~7.3~~	273~~208~~ ± 129~~102~~	**1**	572~~861~~ ± 159~~577~~				293~~640~~ ± 205~~586~~	**0.94~~9~~**		


A
Prior
Halophila ovalis
Amphibolis antarctica
Seagrasses
α (μmol g⁻¹ h⁻¹)
τ (μmol g⁻¹ h⁻¹)
k (d⁻¹)
μ (d)
Pmax (μmol O2 g⁻¹ h⁻¹)
Detrital age (d)
B
k (d⁻¹)
μ (d)
Pmax (μmol O2 leaf⁻¹ h⁻¹)
Detrital age (d)


Terrestrial plants
Freshwater plants
Seagrasses
Seaweeds
k (d⁻¹)
μ (d)
P
Chl
Prior
Light
Dark
P (normalised)
Chl (normalised)
Detrital age (d)

Supplement to

Only macroalgal detritus remains viable for months

Luka Seamus Wright[1,2]*
[1]Oceans Institute, University of Western Australia, Perth, Australia
[2]School of Biological Sciences, University of Western Australia, Perth, Australia
*luka.wright@research.uwa.edu.au, luka@wright.it

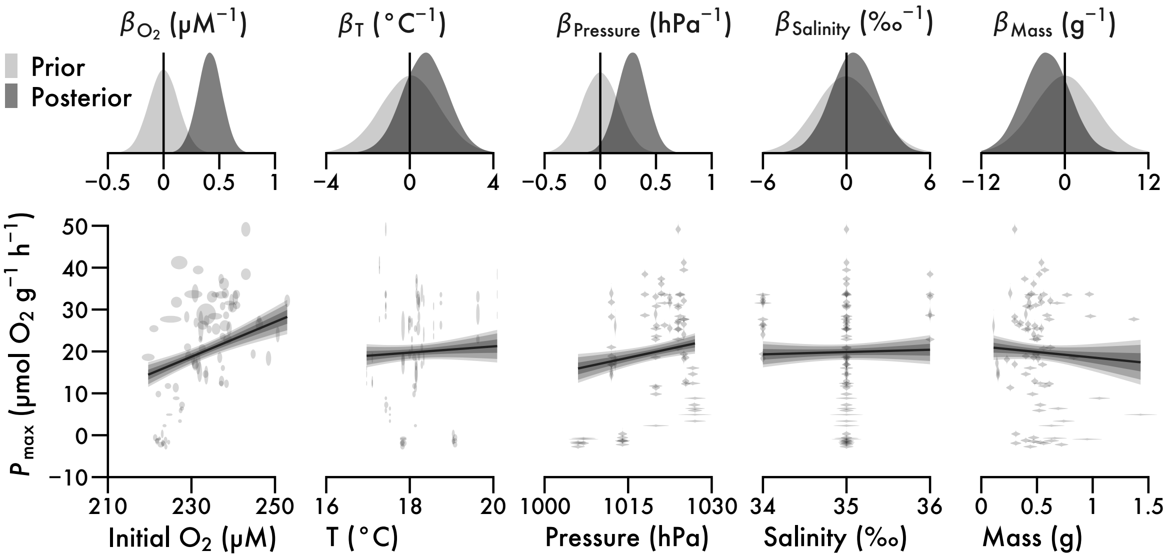


Figure 1. Effect of confounding variables associated with incubation on light-saturated net seagrass photosynthesis (P_{max}) per gram of blotted mass. Distributions are kernel density estimates of priors and posteriors for slopes of the multiple linear regression. P_{max}, initial oxygen (O_2) and temperature (T) are measured with error. Ellipses and violins are bi- and univariate posterior probability distributions for observations. Lines and intervals are means and the central 50, 80 and 90% of posterior probability for the mean prediction of P_{max}.

Table S1. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies and observations in each non-taxonomic group are highlighted in bold.

	Species	Days	Studies	Observations
Chlorophyta				
<u>Ulvales</u>				
Ulvaceae	*Ulva clathrata*	30	1	6
	Ulva intestinalis	30	1	6
	Ulva reticulata	30	1	6
Heterokontophyta				
<u>Desmarestiales</u>				
Desmarestiaceae	*Desmarestia anceps*	**313**	1	65
<u>Fucales</u>				
Durvillaeaceae	*Durvillaea antarctica*	14	2	448
Sargassaceae	*Sargassum fallax*	14	1	84
	Sargassum spinuligerum	14	1	159
Seirococcaceae	*Scytothalia dorycarpa*	14	1	48
<u>Laminariales</u>				

Laminariaceae	*Laminaria digitata*	46	3	162
	Laminaria hyperborea	168	**6**	**533**
	Laminaria ochroleuca	56	4	416
	Macrocystis pyrifera	55	3	363
	Saccharina latissima	39	1	42
Lessoniaceae	*Ecklonia radiata*	119	2	117
<u>Tilopteridales</u>				
Phyllariaceae	*Saccorhiza polyschides*	39	1	38
Streptophyta				
<u>Alismatales</u>				
Cymodoceaceae	*Amphibolis antarctica**	34	**1**	39
Hydrocharitaceae	*Elodea densa*	16	2	43
	*Halophila ovalis**	34	**1**	42
	Hydrilla verticillata	12	**7**	**506**
	*Thalassia testudinum**	52	**1**	14
	Vallisneria americana	**104**	1	5
	Vallisneria spiralis	9	6	59
Potamogetonaceae	*Potamogeton* sp.	**104**	1	5
	Stuckenia pectinata	9	5	55
Zosteraceae	*Zostera marina**	**85**	**1**	10
	*Zostera muelleri**	16	**1**	**58**
<u>Apiales</u>				
Apiaceae	*Coriandrum sativum*	9	1	24
	Petroselinum crispum	7	1	7
<u>Asterales</u>				
Asteraceae	*Chrysanthemum × morifolium*	12	1	30
	Helianthus annuus	0.024	1	36
	Helianthus tuberosus	0.026	1	36
	Inula racemosa	0.025	1	36
	Smallanthus connatus	0.026	1	36
<u>Brassicales</u>				
Brassicaceae	*Arabidopsis thaliana*	6	2	60
	Brassica napus	1.7	1	125
	Brassica oleracea	10	1	22
	Nasturtium officinale	5	2	39
	Raphanus raphanistrum	6	1	20
	Sinapis alba	6	1	20
Tropaeolaceae	*Tropaeolum majus*	8	2	27
<u>Caryophyllales</u>				
Amaranthaceae	*Amaranthus cruentus*	0.026	1	36
	Celosia argentea	0.024	1	36
	Chenopodium berlandieri	0.27	1	26
	Gomphrena serrata	0.024	1	36
	Spinacia oleracea	10	3	85
Polygonaceae	*Koenigia weyrichii*	0.028	1	36
<u>Cucurbitales</u>				
Cucurbitaceae	*Cucumis sativus*	13	2	13
	Sicyos edulis	10	1	48
<u>Ericales</u>				
Theaceae	*Camellia sinensis*	4	1	11
<u>Fabales</u>				
Fabaceae	*Cercis canadensis*	0.0052	1	576

	Lathyrus oleraceus	17	2	42
	Phaseolus vulgaris	2.1	1	10
	Trifolium subterraneum	7	2	25
	Vigna unguiculata	4	1	170
Fagales				
Betulaceae	*Betula pendula*	0.029	1	36
Fagaceae	*Quercus alba*	0.01	1	24
	Quercus muehlenbergii	0.0052	1	192
	Quercus robur	0.024	2	228
	Quercus rubra	0.01	1	24
Juglandaceae	*Juglans nigra*	0.01	1	24
Lamiales				
Lamiaceae	*Salvia officinalis*	11	1	8
Laurales				
Lauraceae	*Cinnamomum tamala*	21	1	36
Pinales				
Pinaceae	*Larix sibirica*	0.025	1	36
	Pinus sibirica	0.023	1	33
	Pinus sylvestris	0.022	1	33
Poales				
Cyperaceae	*Bolboschoenus fluviatilis*	**104**	1	5
Poaceae	*Avena sativa*	8	8	161
	Hordeum vulgare	11	**13**	466
	Lolium multiflorum	5	1	44
	Lolium pratense	7	5	112
	Oryza sativa	27	11	327
	Panicum miliaceum	5	2	69
	Secale cereale	5	2	102
	Triticum aestivum	10	**13**	646
	Triticum turgidum	6	1	7
	Triticum vulgare	6	1	48
	Zea mays	6	3	58
Typhaceae	*Typha angustifolia*	**104**	1	5
Polypodiales				
Nephrolepidaceae	*Nephrolepis exaltata*	10	1	24
Rosales				
Rosaceae	*Fragaria × ananassa*	0.0056	1	663
	Malus domestica	21	3	150
	Malus hupehensis	15	1	18
Sapindales				
Rutaceae	*Citrus × aurantium*	10	1	5
Sapindaceae	*Acer saccharinum*	0.25	1	36
	Acer truncatum	0.0052	1	192
Solanales				
Convolvulaceae	*Ipomoea batatas*	3	1	24
Solanaceae	*Nicotiana rustica*	10	1	180
	Nicotiana tabacum	9	1	24
	Solanum lycopersicum	5	1	192
	Solanum melongena	16	1	24
	Solanum tuberosum	0.026	1	36
Vitales				
Vitaceae	*Vitis vinifera*	0.0052	1	**1280**

References. Studies included in the meta-analysis in addition to the present study. Note that 128 references are listed because the data from three studies were not independent and thus merged.

Albright LJ, Chocair J, Masuda K, Valdes M. **1982**. Degradation of the kelps *Macrocystis integrifolia* and *Nereocystis luetkeana* in British Columbia coastal waters. In Srivastava LM (ed) Synthetic and Degradative Processes in Marine Macrophytes. De Gruyter, Berlin, pp 215-234. https://doi.org/10.1515/9783110837988-015.

Arguelles T, Guardiola JL. **1977**. Hormonal control of senescence in excised orange leaves. *Journal of Horticultural Science* **52**: 199–204. https://doi.org/10.1080/00221589.1977.11514748.

Asim M, Guo M, Khan R *et al*. **2022**. Investigation of sugar signaling behaviors involved in sucrose-induced senescence initiation and progression in *N. tabacum*. *Plant Physiology and Biochemistry* **184**: 112–125. https://doi.org/10.1016/j.plaphy.2022.05.024.

Barua DN. **1960**. Effect of age and carbon-dioxide concentration on assimilation by detached leaves of tea and sunflower. *The Journal of Agricultural Science* **55**: 413–421. https://doi.org/10.1017/s0021859600023327.

de Bettignies F, Dauby P, Thomas F *et al*. **2020**. Degradation dynamics and processes associated with the accumulation of *Laminaria hyperborea* (Phaeophyceae) kelp fragments: an in situ experimental approach. *Journal of Phycology* **56**: 1481–1492. https://doi.org/10.1111/jpy.13041.

Bianchi TS, Findlay S. **1991**. Decomposition of Hudson estuary macrophytes: photosynthetic pigment transformations and decay constants. *Estuaries* **14**: 65–73. https://doi.org/10.2307/1351983.

Birecka H, Ireton KP, Bitonti AJ, McCann PP. **1991**. Endogenous polyamine levels and dark-induced leaf senescence. *Phytochemistry* **30**: 105–108. https://doi.org/10.1016/0031-9422(91)84107-4.

Biswal B, Joshi PN, Kulandaivelu G. **1997**. Changes in leaf protein and pigment contents and photosynthetic activities during senescence of detached maize leaves: influence of different ultraviolet radiations. *Photosynthetica* **34**: 37–44. https://doi.org/10.1023/a:1006855532424.

Biswal B, NK C, Sahu P, UC B. **1983**. Senescence of detached fern leaves. *Plant and Cell Physiology* **24**: 1203–1208. https://doi.org/10.1093/oxfordjournals.pcp.a076634.

Biswal UC, Mohanty P. **1976a**. Dark stress induced senescence of detached barley leaves: II. Alteration in the absorption characteristic and photochemical activity of the chloroplasts isolated from senescing leaves. *Plant Science Letters* **7**: 371–379. https://doi.org/10.1016/0304-4211(76)90094-8.

Biswal UC, Mohanty P. **1976b**. Aging induced changes in photosynthetic electron transport of detached barley leaves. *Plant and Cell Physiology* **17**: 323–331. https://doi.org/10.1093/oxfordjournals.pcp.a075285.

Biswas AK, Choudhuri MA. **1980**. Mechanism of monocarpic senescence in rice. *Plant Physiology* **65**: 340–345. https://doi.org/10.1104/pp.65.2.340.

Biswas R, Sinhababu A, Banerjee A, Kar RK. **2005**. Qualitative and quantitative effect of light on leaf senescence in a submerged aquatic weed, *Hydrilla verticillata*. *Indian Journal of Plant Physiology* **10**: 66.

Bouma D. **1975**. Effects of some metabolic phosphorus compounds on rates of photosynthesis of detached phosphorus-deficient subterranean clover leaves. *Journal of Experimental Botany* **26**: 52–59. https://doi.org/10.1093/jxb/26.1.52.

Bouma D, Dowling EJ. **1982**. The relationship between the phosphorus status of subterranean clover plants and losses in chlorophyll from detached leaves: a basis for the visual diagnosis of phosphorus deficiency. *Annals of Botany* **49**: 637–648. https://doi.org/10.1093/oxfordjournals.aob.a086291.

Brady CJ, Heng FT. **1975**. Rate of protein synthesis in senescing, detached wheat leaves. *Functional Plant Biology* **2**: 163–176. https://doi.org/10.1071/pp9750163.

Brouwer PEM. **1996**. Decomposition in situ of the sublittoral Antarctic macroalga *Desmarestia anceps* Montagne. *Polar Biology* **16**: 129–137. https://doi.org/10.1007/bf02390433.

Causin HF, Marchetti CF, Pena LB, Gallego SM, Barneix AJ. **2015**. Down-regulation of catalase activity contributes to senescence induction in wheat leaves exposed to shading stress. *Biologia Plantarum* **59**: 154–162. https://doi.org/10.1007/s10535-014-0480-z.

Chalise R, Dahal A, Basnet S, Sharma S, Pant DR, Khanal R. **2024**. Effect of plasma-activated water on chlorophyll retention in detached Tejpat (*Cinnamomum tamala*) leaves. *Heliyon* **10**: e24480. https://doi.org/10.1016/j.heliyon.2024.e24480.

Chen H-J, Tsai Y-J, Chen W-S, Huang G-J, Huang S-S, Lin Y-H. **2010**. Ethephon-mediated effects on leaf senescence are affected by reduced glutathione and EGTA in sweet potato detached leaves. *Botanical Studies* **51**: 171–181.

Claussen W, Hawker JS, Loveys BR. **1985**. Sucrose synthase activity, invertase activity, net photosynthetic rates and carbohydrate content of detached leaves of eggplants as affected by attached stems and shoots (sinks). *Journal of Plant Physiology* **119**: 123–131. https://doi.org/10.1016/s0176-1617(85)80171-1.

Dent RM, Cowan AK. **1994**. Abscisic acid-induced senescence in light-and dark-incubated detached leaves of barley. *South African Journal of Botany* **60**: 340–344. https://doi.org/10.1016/s0254-6299(16)30589-0.

Dolui TK, Jana S. **1988**. Effects of phytohormones on some biochemical parameters during dark induced leaf senescence of *Sechium edule* on Darjeeling hill of the Eastern Himalayas. *Biologia Plantarum* **30**: 379–383. https://doi.org/10.1007/bf02878194.

Embry JL, Nothnagel EA. **1994**. Photosynthetic light-harvesting during leaf senescence in *Panicum miliaceum*. *Plant Science* **95**: 141–152. https://doi.org/10.1016/0168-9452(94)90088-4.

Embry JL, Nothnagel EA. **1988**. Leaf development and senescence in *Panicum miliaceum* L., a cereal with a short seed-to-seed cycle. *Plant Science* **55**: 129–136. https://doi.org/10.1016/0168-9452(88)90168-9.

Even-Chen Z, Atsmon D, Itai C. **1978**. Hormonal aspects of senescence in detached tobacco leaves. *Physiologia Plantarum* **44**: 377–382. https://doi.org/10.1111/j.1399-3054.1978.tb01641.x.

Frontier N, de Bettignies F, Foggo A, Davoult D. **2021**. Sustained productivity and respiration of degrading kelp detritus in the shallow benthos: detached or broken, but not dead. *Marine Environmental Research* **166**: 105277. https://doi.org/10.1016/j.marenvres.2021.105277.

Frontier N, Mulas M, Foggo A, Smale DA. **2022**. The influence of light and temperature on detritus degradation rates for kelp species with contrasting thermal affinities. *Marine Environmental Research* **173**: 105529. https://doi.org/10.1016/j.marenvres.2021.105529.

Gapper NE, Coupe SA, McKenzie MJ *et al.* **2005**. Senescence-associated down-regulation of 1-aminocyclopropane-1-carboxylate (ACC) oxidase delays harvest-induced senescence in broccoli. *Functional Plant Biology* **32**: 891–901. https://doi.org/10.1071/fp05076.

Gauthier M-M, Jacobs DF. **2018**. Reductions in net photosynthesis and stomatal conductance vary with time since leaf detachment in three deciduous angiosperms. *Trees* **32**: 1247–1252. https://doi.org/10.1007/s00468-018-1706-z.

Gi E, Cho S-H, Kim S-H, Kang K, Paek N-C. **2024**. Rice ONAC016 promotes leaf senescence through abscisic acid signaling pathway involving OsNAP. *The Crop Journal* https://doi.org/10.1016/j.cj.2024.02.009.

Grover A, Sabat SC, Mohanty P. **1986**. Effect of temperature on photosynthetic activities of senescing detached wheat leaves. *Plant and Cell Physiology* **27**: 117–126. https://doi.org/10.1093/oxfordjournals.pcp.a077071.

Grover A, Sabat SC, Mohanty P. **1987**. Does the loss of leaf chlorophyll during senescence of primary wheat leaf arise due to loss in chloroplast number or chlorophyll content. *Biochemie und Physiologie der Pflanzen* **182**: 481–484. https://doi.org/10.1016/s0015-3796(87)80078-1.

Gut H, Rutz C, Matile P, Thomas H. **1987**. Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: degradation of carotenoids. *Physiologia Plantarum* **70**: 659–663. https://doi.org/10.1111/j.1399-3054.1987.tb04321.x.

Hilditch P, Thomas H, Rogers L. **1986**. Leaf senescence in a non-yellowing mutant of *Festuca pratensis*: photosynthesis and photosynthetic electron transport. *Planta* **167**: 146–151. https://doi.org/10.1007/bf00446382.

Housley TL, Pollock CJ. **1985**. Photosynthesis and carbohydrate metabolism in detached leaves of *Lolium temulentum* L. *New Phytologist* **99**: 499–507. https://doi.org/10.1111/j.1469-8137.1985.tb03678.x.

Huang J, Han B, Xu S, Zhou M, Shen W. **2011**. Heme oxygenase-1 is involved in the cytokinin-induced alleviation of senescence in detached wheat leaves during dark incubation. *Journal of Plant Physiology* **168**: 768–775. https://doi.org/10.1016/j.jplph.2010.10.010.

Huang M, Feng L, He Y, Zhu Z. **2006**. Study on changing rules of chlorophyll concentration of detached canola leaves. In Xu K, Luo Q, Xing D, Priezzhev AV, Tuchin VV (eds) Fourth International Conference on Photonics and Imaging in Biology and Medicine, 6047. SPIE, Tianjin, pp 604740. https://doi.org/10.1117/12.710949.

Hukmani P, Tripathy BC. **1994**. Chlorophyll biosynthetic reactions during senescence of excised barley (*Hordeum vulgare* L. cv IB 65) leaves. *Plant Physiology* **105**: 1295–1300. https://doi.org/10.1104/pp.105.4.1295.

Jana S, Choudhuri MA. **1979**. Photosynthetic, photorespiratory and respiratory behaviour of three submersed aquatic angiosperms. *Aquatic Botany* **7**: 13–19. https://doi.org/10.1016/0304-3770(79)90003-2.

Jana S, Choudhuri MA. **1980**. Senescence in submerged aquatic angiosperms: changes in intact and isolated leaves during aging. *New Phytologist* **86**: 191–198. https://doi.org/10.1111/j.1469-8137.1980.tb03188.x.

Jana S, Choudhuri MA. **1982a**. Changes occurring during aging and senescence in a submerged aquatic angiosperm (*Potamogeton pectinatus*). *Physiologia Plantarum* **55**: 356–360. https://doi.org/10.1111/j.1399-3054.1982.tb00305.x.

Jana S, Choudhuri MA. **1982b**. Glycolate metabolism of three submersed aquatic angiosperms during ageing. *Aquatic Botany* **12**: 345–354. https://doi.org/10.1016/0304-3770(82)90026-2.

Jana S, Choudhuri MA. **1982c**. Changes in the activities of ribulose 1, 5-bisphosphate and phosphoenolpyruvate carboxylases in submersed aquatic angiosperms during aging. *Plant Physiology* **70**: 1125–1127. https://doi.org/10.1104/pp.70.4.1125.

Jana S, Choudhuri MA. **1984a**. Synergistic effects of heavy metal pollutants on senescence in submerged aquatic plants. *Water, Air, & Soil Pollution* **21**: 351–357. https://doi.org/10.1007/bf00163635.

Jana S, Choudhuri MA. **1984b**. Effects of plant growth regulators on Hill activity of submerged aquatic plants during induced senescence. *Aquatic Botany* **18**: 371–376. https://doi.org/10.1016/0304-3770(84)90057-3.

Jana S. **1988**. Glycollate metabolism of wheat and rice leaves during senescence and under the influence of growth regulators. *Biologia Plantarum* **30**: 30–33. https://doi.org/10.1007/bf02876420.

Jia HS, Han YQ, Li DQ. **2003**. Photoinhibition and active oxygen species production in detached apple leaves during dehydration. *Photosynthetica* **41**: 151–156. https://doi.org/10.1023/a:1025889219115.

Jiang W, Sheng Q, Zhou X-J, Zhang M-J, Liu X-J. **2002**. Regulation of detached coriander leaf senescence by 1-methylcyclopropene and ethylene. *Postharvest Biology and Technology* **26**: 339–345. https://doi.org/10.1016/s0925-5214(02)00068-6.

Kang K, Kim YS, Park S, Back K. **2009**. Evaluation of light-harvesting complex proteins as senescence-related protein markers in detached rice leaves. *Photosynthetica* **47**: 638–640. https://doi.org/10.1007/s11099-009-0093-5.

Kao CH, Yang SF. **1983**. Role of ethylene in the senescence of detached rice leaves. *Plant Physiology* **73**: 881–885. https://doi.org/10.1104/pp.73.4.881.

Kar M, Feierabend J. **1984**. Metabolism of activated oxygen in detached wheat and rye leaves and its relevance to the initiation of senescence. *Planta* **160**: 385–391. https://doi.org/10.1007/bf00429753.

Kar M, Streb P, Hertwig B, Feierabend J. **1993**. Sensitivity to photodamage increases during senescence in excised leaves. *Journal of Plant Physiology* **141**: 538–544. https://doi.org/10.1016/s0176-1617(11)80453-0.

Kar RK, Choudhuri MA. **1985**. Senescence of *Hydrilla* leaves in light and darkness. *Physiologia Plantarum* **63**: 225–230. https://doi.org/10.1111/j.1399-3054.1985.tb01907.x.

Kar S, Montague DT, Villanueva-Morales A. **2021a**. Measurement of photosynthesis in excised leaves of ornamental trees: a novel method to estimate leaf level drought tolerance and increase experimental sample size. *Trees* **35**: 889–905. https://doi.org/10.1007/s00468-021-02088-w.

Kar S, Montague T, Villanueva-Morales A, Hellman E. **2021b**. Measurement of gas exchange on excised grapevine leaves does not differ from *in situ* leaves, and potentially shortens sampling time. *Applied Sciences* **11**: 3644. https://doi.org/10.3390/app11083644.

Karatas I, Ozturk L, Ersahin Y, Okatan Y. **2010**. Effects of auxin on photosynthetic pigments and some enzyme activities during dark-induced senescence of *Tropaeolum* leaves. *Pakistan Journal of Botany* **42**: 1881–1888.

Kato MC, Hikosaka K, Hirose T. **2002**. Leaf discs floated on water are different from intact leaves in photosynthesis and photoinhibition. *Photosynthesis Research* **72**: 65–70. https://doi.org/10.1023/a:1016097312036.

Knauer GA, Ayers AV. **1977**. Changes in carbon, nitrogen, adenosine triphosphate, and chlorophyll *a* in decomposing *Thalassia testudinum* leaves. *Limnology and Oceanography* **22**: 408–414. https://doi.org/10.4319/lo.1977.22.3.0408.

Kreslavski VD, Lankin AV, Vasilyeva GK *et al.* **2014**. Effects of polyaromatic hydrocarbons on photosystem II activity in pea leaves. *Plant Physiology and Biochemistry* **81**: 135–142. https://doi.org/10.1016/j.plaphy.2014.02.020.

Landolt R, Matile P. **1990**. Glyoxisome-like microbodies in senescent spinach leaves. *Plant Science* **72**: 159–163. https://doi.org/10.1016/0168-9452(90)90078-3.

Lewington RJ, Talbot M, Simon EW. **1967**. The yellowing of attached and detached cucumber cotyledons. *Journal of Experimental Botany* **18**: 526–534. https://doi.org/10.1093/jxb/18.3.526.

Lewington RJ, Simon EW. **1969**. The effect of light on the senescence of detached cucumber cotyledons. *Journal of Experimental Botany* **20**: 138–144. https://doi.org/10.1093/jxb/20.1.138.

Liu L, Li H, Zeng H, Cai Q, Zhou X, Yin C. **2016**. Exogenous jasmonic acid and cytokinin antagonistically regulate rice flag leaf senescence by mediating chlorophyll degradation, membrane deterioration, and senescence-associated genes expression. *Journal of Plant Growth Regulation* **35**: 366–376. https://doi.org/10.1007/s00344-015-9539-0.

Lovell PH, Illsley ANN, Moore KG. **1972**. The effects of light intensity and sucrose on root formation, photosynthetic ability, and senescence in detached cotyledons of *Sinapis alba* L.

and *Raphanus sativus* L. *Annals of Botany* **36**: 123–134. https://doi.org/10.1093/oxfordjournals.aob.a084565.

Martin C, Thimann KV. **1972**. The role of protein synthesis in the senescence of leaves: I. The formation of protease. *Plant Physiology* **49**: 64–71. https://doi.org/10.1104/pp.49.1.64.

Miller BL, Huffaker RC. **1985**. Differential induction of endoproteinases during senescence of attached and detached barley leaves. *Plant Physiology* **78**: 442–446. https://doi.org/10.1104/pp.78.3.442.

Mishra D, Kar M. **1973**. Effect of benzimidazole and nickel ions in the control of *Oryza sativa* leaf senescence by red and far-red light. *Phytochemistry* **12**: 1521–1522. https://doi.org/10.1016/0031-9422(73)80360-7.

Misr AN, Meena M. **1986**. Effect of temperature on senescing detached rice leaves: I. Photoelectron transport activity of chloroplasts. *Plant Science* **46**: 1–4. https://doi.org/10.1016/0168-9452(86)90123-8.

Misra AN, Biswal UC. **1980**. Effect of phytohormones on chlorophyll degradation during aging of chloroplasts in vivo and in vitro. *Protoplasma* **105**: 1–8. https://doi.org/10.1007/bf01279845.

Misra AN, Biswal UC. **1982**. Differential changes in the electron transport properties of thylakoid membranes during aging of attached and detached leaves, and of isolated chloroplasts. *Plant, Cell & Environment* **5**: 27–30. https://doi.org/10.1111/1365-3040.ep11587573.

Momose T, Ozeki Y. **2013**. Regulatory effect of stems on sucrose-induced chlorophyll degradation and anthocyanin synthesis in *Egeria densa* leaves. *Journal of Plant Research* **126**: 859–867. https://doi.org/10.1007/s10265-013-0581-3.

Mondal R, Choudhuri MA. **1981**. Role of hydrogen peroxide in senescence of excised leaves of rice and maize. *Biochemie und Physiologie der Pflanzen* **176**: 700–709. https://doi.org/10.1016/s0015-3796(81)80055-8.

Montague T, Villanueva-Morales A, Khan MAR, Wallace R, Panta S. **2024**. Comparing the accuracy and efficiency of leaf gas exchange measurements of excised and nonexcised strawberry (*Fragaria* × *ananassa* Duch. 'Camino Real') leaves. *HortTechnology* **34**: 728–737. https://doi.org/10.21273/horttech05516-24.

Nedunchezhian N, Ravindran KC, Kulandaivelu G. **1995**. Changes in photosynthetic apparatus during dark incubation of detached leaves from control and ultraviolet-B treated *Vigna* seedlings. *Biologia Plantarum* **37**: 341–348. https://doi.org/10.1007/bf02913238.

Nock LP, Rogers LJ, Thomas H. **1992**. Metabolism of protein and chlorophyll in leaf tissue of *Festuca pratensis* during chloroplast assembly and senescence. *Phytochemistry* **31**: 1465–1470. https://doi.org/10.1016/0031-9422(92)83088-g.

Ogweno J-O, Song X-S, Hu W-H, Shi K, Zhou Y-H, Yu J-Q. **2009**. Detached leaves of tomato differ in their photosynthetic physiological response to moderate high and low temperature stress. *Scientia Horticulturae* **123**: 17–22. https://doi.org/10.1016/j.scienta.2009.07.011.

Oh M-H, Moon Y-H, Lee C-H. **2003**. Increased stability of LHCII by aggregate formation during dark-induced leaf senescence in the *Arabidopsis* mutant, ore10. *Plant and Cell Physiology* **44**: 1368–1377. https://doi.org/10.1093/pcp/pcg170.

Okada K, Inoue Y, Satoh K, Katoh S. **1992**. Effects of light on degradation of chlorophyll and proteins during senescence of detached rice leaves. *Plant and Cell Physiology* **33**: 1183–1191. https://doi.org/10.1093/oxfordjournals.pcp.a078372.

Pellikaan GC. **1982**. Decomposition processes of eelgrass, *Zostera marina* L. *Hydrobiological Bulletin* **16**: 83–92. https://doi.org/10.1007/bf02255416.

Philosoph-Hadas S, Meir S, Akiri B, Kanner J. **1994a**. Oxidative defense systems in leaves of three edible herb species in relation to their senescence rates. *Journal of Agricultural and Food Chemistry* **42**: 2376–2381. https://doi.org/10.1021/jf00047a003.

Philosoph-Hadas S, Meir S, Aharoni N. 1991. Effect of wounding on ethylene biosynthesis and senescence of detached spinach leaves. *Physiologia Plantarum* **83**: 241–246. https://doi.org/10.1111/j.1399-3054.1991.tb02148.x.

Philosoph-Hadas S, Meir S, Aharoni N. 1994b. Role of ethylene in senescence of watercress leaves. *Physiologia Plantarum* **90**: 553–559. https://doi.org/10.1111/j.1399-3054.1994.tb08814.x.

Reyes-Arribas T, Barrett JE, Huber DJ, Nell TA, Clark DG. 2001. Leaf senescence in a non-yellowing cultivar of chrysanthemum (*Dendranthema grandiflora*). *Physiologia Plantarum* **111**: 540–544. https://doi.org/10.1034/j.1399-3054.2001.1110415.x.

Richardson SD. 1957. The effect of leaf age on the rate of photosynthesis in detached leaves of tree seedlings. *Acta Botanica Neerlandica* **6**: 445–457. https://doi.org/10.1111/j.1438-8677.1957.tb00592.x.

Rodoni S, Schellenberg M, Matile P. 1998. Chlorophyll breakdown in senescing barley leaves as correlated with phaeophorbidea oxygenase activity. *Journal of Plant Physiology* **152**: 139–144. https://doi.org/10.1016/s0176-1617(98)80124-7.

Rodríguez MT, Gonzélez MP, Linares JM. 1987. Degradation of chlorophyll and chlorophyllase activity in senescing barley leaves. *Journal of Plant Physiology* **129**: 369–374. https://doi.org/10.1016/s0176-1617(87)80094-9.

Rothäusler E, Gómez I, Hinojosa IA *et al.* **2011**. Kelp rafts in the Humboldt Current: Interplay of abiotic and biotic factors limit their floating persistence and dispersal potential. *Limnology and Oceanography* **56**: 1751–1763. https://doi.org/10.4319/lo.2011.56.5.1751.

Sabater B, Rodríguez MT. 1978. Control of chlorophyll degradation in detached leaves of barley and oat through effect of kinetin on chlorophyllase levels. *Physiologia Plantarum* **43**: 274–276. https://doi.org/10.1111/j.1399-3054.1978.tb02577.x.

Sağlam-Çağ S. 2007. The effect of epibrassinolide on senescence in wheat leaves. *Biotechnology & Biotechnological Equipment* **21**: 63–65. https://doi.org/10.1080/13102818.2007.10817415.

Shafique S, Siddiqui PJS, Aziz RA, Burhan Z, Mansoor SN. 2010. Weight loss and changes in organic, inorganic and chlorophyll contents in three species of seaweeds during decomposition. *Pakistan Journal of Botany* **42**: 2599–2604.

Sharma R, Biswal UC. 1976. Effect of kinetin and light on peroxidase activity of detached barley leaves. *Zeitschrift für Pflanzenphysiologie* **78**: 169–172. https://doi.org/10.1016/s0044-328x(78)80188-3.

Shaw M, Bhattacharya PK, Quick WA. 1965. Chlorophyll, protein, and nucleic acid levels in detached, senescing wheat leaves. *Canadian Journal of Botany* **43**: 739–746. https://doi.org/10.1139/b65-083.

Singh N, Mishra D. 1975. The effect of benzimidazole and red light on the senescence of detached leaves of *Oryza sativa* cv. Ratna. *Physiologia Plantarum* **34**: 67–74. https://doi.org/10.1111/j.1399-3054.1975.tb01858.x.

Špundová M, Strzałka K, Nauš J. 2005. Xanthophyll cycle activity in detached barley leaves senescing under dark and light. *Photosynthetica* **43**: 117–124. https://doi.org/10.1007/s11099-005-7124-7.

Špundová M, Popelková H, Ilík P, Skotnica J, Novotný R, Nauš J. 2003. Ultra-structural and functional changes in the chloroplasts of detached barley leaves senescing under dark and light conditions. *Journal of Plant Physiology* **160**: 1051–1058. https://doi.org/10.1078/0176-1617-00902.

Srivastava BI, Ware G. 1965. The effect of kinetin on nucleic acids and nucleases of excised barley leaves. *Plant Physiology* **40**: 62. https://doi.org/10.1104/pp.40.1.62.

Strother S, Vatta R. 1986. The effects of light, cycloheximide and incubation medium on the senescence of detached seagrass leaves. *Biologia Plantarum* **28**: 211–218. https://doi.org/10.1007/bf02894599.

Subhan D, Murthy SDS. 2000. Synergistic effect of AlCl 3 and kinetin on chlorophyll and protein contents and photochemical activities in detached wheat primary leaves during dark incubation. *Photosynthetica* **38**: 211–214. https://doi.org/10.1023/a:1007261614200.
Subhan D, Murthy SDS. 2001a. Senescence retarding effect of metal ions: pigment and protein contents and photochemical activities of detached primary leaves of wheat. *Photosynthetica* **39**: 53–58. https://doi.org/10.1023/a:1012487718114.
Subhan D, Murthy SDS. 2001b. Effect of polyamines on chlorophyll and protein contents, photochemical activity, and energy transfer in detached wheat leaves during dark incubation. *Biologia Plantarum* **44**: 529–533. https://doi.org/10.1023/a:1013734502112.
Tala F, Velásquez M, Mansilla A, Macaya EC, Thiel M. 2016. Latitudinal and seasonal effects on short-term acclimation of floating kelp species from the South-East Pacific. *Journal of Experimental Marine Biology and Ecology* **483**: 31–41. https://doi.org/10.1016/j.jembe.2016.06.003.
Tala F, Penna-Díaz MA, Luna-Jorquera G, Rothäusler E, Thiel M. 2017. Daily and seasonal changes of photobiological responses in floating bull kelp *Durvillaea antarctica* (Chamisso) Hariot (Fucales: Phaeophyceae). *Phycologia* **56**: 271–283. https://doi.org/10.2216/16-93.1.
Tetley RM, Thimann KV. 1974. The metabolism of oat leaves during senescence: I. Respiration, carbohydrate metabolism, and the action of cytokinins. *Plant Physiology* **54**: 294–303. https://doi.org/10.1104/pp.54.3.294.
Thimann KV, Tetley RR, Van Thanh T. 1974. The metabolism of oat leaves during senescence: II. Senescence in leaves attached to the plant. *Plant Physiology* **54**: 859–862. https://doi.org/10.1104/pp.54.6.859.
Thimann KV, Tetley RM, Krivak BM. 1977. Metabolism of oat leaves during senescence: V. Senescence in light. *Plant Physiology* **59**: 448–454. https://doi.org/10.1104/pp.59.3.448.
Thimann KV, Satler SO. 1979. Relation between leaf senescence and stomatal closure: senescence in light. *Proceedings of the National Academy of Sciences* **76**: 2295–2298. https://doi.org/10.1073/pnas.76.5.2295.
Thimann KV. 1985. The senescence of detached leaves of *Tropaeolum*. *Plant Physiology* **79**: 1107–1110. https://doi.org/10.1104/pp.79.4.1107.
Thomas H. 1987. Sid: a Mendelian locus controlling thylakoid membrane disassembly in senescing leaves of *Festuca pratensis*. *Theoretical and Applied Genetics* **73**: 551–555. https://doi.org/10.1007/bf00289193.
Thomas H, Stoddart JL. 1975. Separation of chlorophyll degradation from other senescence processes in leaves of a mutant genotype of meadow fescue (*Festuca pratensis* L.). *Plant Physiology* **56**: 438–441. https://doi.org/10.1104/pp.56.3.438.
Turner WB, Bidwell RGS. 1965. Rates of photosynthesis in attached and detached bean leaves, and the effect of spraying with indoleacetic acid solution. *Plant Physiology* **40**: 446. https://doi.org/10.1104/pp.40.3.446.
van Hees DH, Olsen YS, Mattio L, Ruiz-Montoya L, Wernberg T, Kendrick GA. 2019. Cast adrift: physiology and dispersal of benthic *Sargassum spinuligerum* in surface rafts. *Limnology and Oceanography* **64**: 526–540. 10.1002/lno.11057.
Veierskov B, Kirk HG. 1986. Senescence in oat leaf segments under hypobaric conditions. *Physiologia Plantarum* **66**: 283–287. https://doi.org/10.1111/j.1399-3054.1986.tb02421.x.
Velerskov B. 1987. Irradiance-dependent senescence of isolated leaves. *Physiologia Plantarum* **71**: 316–320. https://doi.org/10.1111/j.1399-3054.1987.tb04349.x.
Vlčková A, Špundová M, Kotabová E, Novotný R, Doležal K, Nauš J. 2006. Protective cytokinin action switches to damaging during senescence of detached wheat leaves in continuous light. *Physiologia Plantarum* **126**: 257–267. https://doi.org/10.1111/j.1399-3054.2006.00593.x.

Voronin PY, Fedoseeva GP. 2012. Stomatal control of photosynthesis in detached leaves of woody and herbaceous plants. *Russian Journal of Plant Physiology* **59**: 281–286. https://doi.org/10.1134/s1021443712020173.

Wang P, Yin L, Liang D, Li C, Ma F, Yue Z. 2012. Delayed senescence of apple leaves by exogenous melatonin treatment: toward regulating the ascorbate-glutathione cycle. *Journal of Pineal Research* **53**: 11–20. https://doi.org/10.1111/j.1600-079x.2011.00966.x.

Wen B, Li C, Fu X *et al.* **2019**. Effects of nitrate deficiency on nitrate assimilation and chlorophyll synthesis of detached apple leaves. *Plant Physiology and Biochemistry* **142**: 363–371. https://doi.org/10.1016/j.plaphy.2019.07.007.

Wright LS, Foggo A. 2021. Photosynthetic pigments of co-occurring Northeast Atlantic *Laminaria* spp. are unaffected by decomposition. *Marine Ecology Progress Series* **678**: 227–232. https://doi.org/10.3354/meps13886.

Wright LS, Pessarrodona A, Foggo A. 2022. Climate-driven shifts in kelp forest composition reduce carbon sequestration potential. *Global Change Biology* **28**: 5514–5531. https://doi.org/10.1111/gcb.16299.

Wright LS, Kregting L. 2023. Genus-specific response of kelp photosynthetic pigments to decomposition. *Marine Biology* **170**: 144. https://doi.org/10.1007/s00227-023-04289-y.

Wright LS, Simpkins T, Filbee-Dexter K, Wernberg T. 2024. Temperature sensitivity of detrital photosynthesis. *Annals of Botany* **133**: 17–28. https://doi.org/10.1093/aob/mcad167.

Xiao X, de Bettignies T, Olsen YS, Agusti S, Duarte CM, Wernberg T. 2015. Sensitivity and acclimation of three canopy-forming seaweeds to UVB radiation and warming. *PLoS One* **10**: e0143031. https://doi.org/10.1371/journal.pone.0143031.

Yang J, Worley E, Udvardi M. 2014. A NAP-AAO3 regulatory module promotes chlorophyll degradation via ABA biosynthesis in *Arabidopsis* leaves. *The Plant Cell* **26**: 4862–4874. https://doi.org/10.1105/tpc.114.133769.

Yoshida Y. 1970. Control of chloroplast senescence of detached *Elodea* leaves. *Shokubutsugaku Zasshi* **83**: 137–143. https://doi.org/10.15281/jplantres1887.83.137.

Zhang Z, Li G, Gao H *et al.* **2012**. Characterization of photosynthetic performance during senescence in stay-green and quick-leaf-senescence *Zea mays* L. inbred lines. *PLoS One* **7**: e42936. https://doi.org/10.1371/journal.pone.0042936.

Zhang Z, Guo J, Yang C *et al.* **2023**. Exogenous dopamine delays the senescence of detached *Malus hupehensis* leaves by reducing phytohormone signalling and sugar degradation. *Scientia Horticulturae* **319**: 112151. https://doi.org/10.1016/j.scienta.2023.112151.

Zhao L, Zhang H, Zhang B, Bai X, Zhou C. 2012. Physiological and molecular changes of detached wheat leaves in responding to various treatments. *Journal of Integrative Plant Biology* **54**: 567–576. https://doi.org/10.1111/j.1744-7909.2012.01139.x.